

Inelastic Capital*

Matthieu Gomez[†]

Émilien Gouin-Bonenfant[‡]

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Abstract

How does the economy absorb a rise in desired saving? In the standard one-sector model, the long-run supply of capital is perfectly elastic: saving shocks raise capital quantities but not asset prices. We show that this result breaks down whenever (i) capital formation is relatively labor-intensive or (ii) there is comparative advantage in the labor market. Using a sufficient statistic approach and US input–output data, we find that roughly 30% of long-run demand shocks are absorbed by higher capital prices, rather than more capital accumulation. The reason is that investment in the modern US economy disproportionately requires specialized labor (e.g., engineers, researchers, managers), which means that skills become a bottleneck that limits capital accumulation. One implication is that capital tax cuts disproportionately benefit those involved in capital formation: wage gains are largest for executives and engineers and smallest for nurses and teachers, and cash-flow gains are largest for owners of land and structures and smallest for owners of intangible capital.

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[†]Columbia University, mg3901@columbia.edu.

[‡]Columbia University, eg3041@columbia.edu.

Introduction

The neoclassical growth model treats capital formation as a frictionless process: a single final good can be used either for consumption or to accumulate capital. A well-understood implication is that the supply curve for capital goods is perfectly elastic (i.e., the relative price of capital goods is pinned down at one). As a result, shocks in the capital market (e.g., changes in preferences, taxes, etc) are entirely absorbed by real investment, without any valuation effects.

This prediction, however, is at odds with the data, where capital asset prices are much more volatile than quantities, even at long horizons. For instance, the global savings glut of the 2000s (see, e.g., the role of international capital flows [Bernanke, 2007](#); demographics [Auclert et al., 2021](#); and rising inequalities [Mian et al., 2024](#)) had little effect on capital formation in the US, but instead led to large revaluations. The q-theory of investment slows down the response of investment, but only in the short run.

This paper takes a different route: we emphasize the fact that the production of capital requires specialized labor (scientists, managers, financiers) whose aggregate supply is inherently limited. For instance, roughly half of aggregate investment in the US is now accounted for by the production of “intellectual property” and “organizational services”, both of which are high-wage, labor-intensive sectors. Even traditional investment sectors such as “construction” are much more labor-intensive than the economy as a whole. In both cases, scaling up aggregate investment implies a rise in the marginal cost of capital.

Using a sectoral production model, we show that this labor bottleneck in capital formation generates an upward-sloping supply curve for capital, even in the long run. What this means is that, generically, shocks in the capital market are absorbed in part by prices, and in part by quantities. Formally, we calculate comparative statics formulas for the response of investment and relative prices to a demand shock (i.e., a shift in capital valuations that leaves technology unchanged). We then map those formulas to US input-output data using a sufficient statistic approach. We conclude that the world lies between the neoclassical growth model (perfectly elastic capital) and an endowment economy (inelastic capital): capital prices more than fully absorb a demand shock on impact and still absorb roughly 30% in the long run, depending on the capital type considered.

Model. In Section 1, we start by calculating the long-run effect of a capital income tax shock on the capital stock as well as the price of capital relative to consumption. We start with a two-sector model (consumption and investment) with two factors of production (labor and capital) in the spirit of [Uzawa \(1961\)](#). Production is Cobb-Douglas and labor supply is CES. We show that whenever (i) investment requires more labor than consumption or (ii) labor is imperfectly mobile across sectors, the cost of producing capital goods *rises* with aggregate investment (i.e., the capital supply curve is upward sloping). Intuitively, as the production of investment goods increases, it puts upward pressure on the price of the factor it uses most intensively. As a result, factor prices

endogenously move against investment good producers, which dampens capital accumulation.

Two sets of sufficient statistics govern the response of quantities and prices to demand shocks: (i) sectoral factor shares, (ii) the labor supply elasticity across sectors. We extend the theory to CES production, multiple capital types, dynamic occupation mobility, adjustment costs, imported inputs, and modified Euler equations. In each case we describe which additional information is needed to quantify the model (for instance how factor shares respond to shocks when moving away from Cobb-Douglas).

Measurement. In Section 2, we use data from the national accounts to study the factor intensity of the consumption and investment goods sectors. In a companion paper (Gomez and Gouin-Bonenfant, 2025), we combine sectoral production data from the BEA’s make-use matrices with employment and compensation data from the Current Population Survey to construct a consolidated sectoral dataset. We allocate 100% of aggregate value added to three broad sectors: consumption, tangible investment, and intangible investment. We then calculate the labor embedded, both directly and indirectly, in sectoral output and decompose it by education level (i.e., high school, college, post-college).

For the purpose of this paper, two important facts emerge. First, the labor share has always been higher in the investment sector, and the divergence has increased over time. Even over the 1970–1990 period, where intangible capital investment was more limited, the fact that the construction sector is very labor-intensive made aggregate investment more labor-intensive than aggregate consumption. As the importance of R&D and the management sector grew, this total labor content gap widened. Second, we highlight the fact that intangible-capital producing sectors disproportionately employ high-education and high-wage workers.

Quantitative predictions. In Section 3, we bring the model to the data in order to quantify the propagation of demand shocks in a quantitative extension of the stylized model. We add three ingredients: (i) multiple types of labor and capital, (ii) occupation-specific labor mobility, (iii) type-specific internal adjustment costs. For the calibration of sectoral technology in the model, we use our sectoral dataset combined with assumptions on the link between sectoral ownership of capital and sectoral use of capital services. For the calibration of the labor supply system, we estimate occupation transition matrices and cross-elasticities from the CPS, combined with external evidence on education choice.

The calibrated model predicts a low aggregate capital supply elasticity of roughly 2 at a ten-year horizon and 5 in the long run. This low level of elasticity has first-order implications on the incidence of macro shocks, cutting the quantity effect of a given demand shock by roughly half at a ten-year horizon (and 20% in the long run) relative to the neoclassical growth model. We decompose the long-run price effect into technology asymmetry, labor specialization, and capital-type composition terms.

Finally, we use the calibrated model to quantify the incidence of corporate taxes in our economy. In the neoclassical growth model, capital tax cuts have a large effect on investment, and their benefits are captured mostly by workers via higher wages. Instead, our model predicts that a substantially larger share of the incidence of a permanent tax cut is borne by capitalists (via revaluation effects) and investment-sector workers (via higher wages). The real effect of the tax cut is modest. We validate this prediction—the split of nominal investment responses into prices and quantities—against quasi-experimental evidence on investment tax incentives (Goolsbee, 1998; House and Shapiro, 2008; House et al., 2022). Because capital barely responds, the efficiency gains from cutting capital taxes are small even at high initial tax rates.

Literature review. This paper revisits Uzawa (1961)’s insight that macroeconomic dynamics depend on technology asymmetry in the production of consumption versus investment. We consider the minimal departure from his model that can account for US data (mainly multiple types of capital and labor and imperfect labor mobility) and calculate the effect of demand shocks in the capital market on quantities and prices. This is a central question in medium-run macro and finance: whether we live in an inelastic economy (Lucas’ orchard) or an infinitely-elastic one (neoclassical growth model) has radically different implications for welfare incidence and optimal policy (more on this shortly). A large literature interprets movements in the relative price of investment goods as investment-specific technology shocks (Greenwood et al., 1997; Justiniano et al., 2011). In our model, demand shocks also move the relative price of investment goods, along an upward-sloping supply curve.

The core insight is that sectoral technology differences dampen the response of real investment in response to a demand shock, even in the long run. Our contribution is to map the theory to the data using a sufficient statistic approach and generate quantitative predictions. Our measurement exercise builds on an accounting framework described in detail in a companion paper (Gomez and Gouin-Bonenfant, 2025) (US data over 1963–2023). We document the fact that investment is more labor-intensive than consumption, and increasingly so with the rising importance of R&D and organizational capital in investment. Earlier empirical work that measures the difference in the labor share in the consumption sector versus the investment sector include Chari et al. (1996), Cummins and Violante (2002) and Valentinyi and Herrendorf (2008) — our key contribution is to extend the definition of investment goods to intangibles, including organization capital (e.g., Eisfeldt and Papanikolaou, 2013; Peters and Taylor, 2017; Eisfeldt, Falato and Xiaolan, 2023; Corrado, Haskel, Jona-Lasinio and Iommi, 2022).¹ Note that we aggregate economic activity in broad expenditure sectors (consumption versus investment) rather than measuring the full production

¹Intangible capital is thought to have distinct properties, relative to structures and equipment. In Eisfeldt and Papanikolaou (2013), intangible capital is unique because it is embedded in key talents. In Crouzet, Eberly, Eisfeldt and Papanikolaou (2022), intangible capital tends to be non-rival and is characterized by limited excludability, which prevents firms from capturing all rents. We instead focus on a distinct input in the production of intangible capital: it requires specialized labor in finite supply.

and investment network at the industry-level (as in Vom Lehn and Winberry, 2022; Casal and Caunedo, 2024; Baqaee and Malmberg, 2025). The reason is that we are interested in demand shocks (which lift the demand for capital goods), rather than industry-level productivity shocks (which reallocate inputs between industries).

Finally, our theory is *not* about a disconnect between investment and q (see, e.g., Gutiérrez and Philippon, 2017; Farhi and Gourio, 2018; Crouzet and Eberly, 2023), but rather an inelastic supply of capital goods, which dampens the real effect of demand shocks. Our analysis focuses on Walrasian forces (i.e., price adjustments and sectoral reallocation of finite resources). As discussed in Mussa (1977), our core channel, which works via equilibrium price adjustments, differs from models with capital adjustment cost functions (see, e.g., Lucas Jr, 1967; Uzawa, 1969; Hayashi, 1982).

1 Theory

We study a simple two-sector model with consumption and investment goods. We derive a sufficient-statistic formula for the long-run elasticity of capital prices and quantity to shocks in the demand for savings. Two forces determine long-run effects: the difference in the capital shares between sectors (which governs the ability to scale the technology), and the structure of comparative advantage in the labor market (which governs how easy it is to reallocate labor across sectors).

1.1 Stylized model

Production in each sector requires a combination of labor and capital. The model differs from the benchmark neoclassical growth model in two ways: the consumption and investment sectors have heterogeneous factor intensities and the sectoral labor supply is imperfectly elastic. We introduce these two ingredients because they capture the main forces omitted by the standard one-sector model that make the long-run supply of capital less than perfectly elastic.

Supply side. We consider an infinite-horizon economy populated by representative firms and households. We suppress time subscripts for clarity. There are two factors, capital K and labor L , and two final goods, C and I . Production in each sector is Cobb-Douglas with constant returns to scale:

$$\begin{aligned} C &= A_C K_C^{\alpha_C} L_C^{1-\alpha_C}, & \alpha_C &\in (0, 1), & \text{(production- } C \text{ sector)} \\ I &= A_I K_I^{\alpha_I} L_I^{1-\alpha_I}, & \alpha_I &\in (0, 1). & \text{(production- } I \text{ sector)} \end{aligned}$$

Notice that the usual one-sector neoclassical growth model technology is nested when $\alpha_C = \alpha_I$. Empirically, we will show in the next section that the consumption sector has a higher capital

intensity than the investment sector; that is, $\alpha_C > \alpha_I$. Investment goods are accumulated into the total stock of capital K , which depreciates at rate δ :

$$K' = (1 - \delta)K + I, \quad \delta \in (0, 1). \quad (\text{capital accumulation})$$

Demand side. Households have CRRA utility with elasticity of intertemporal substitution ψ and impatience parameter β . Consumption is the numeraire and we denote by p_K the price of the investment good relative to the consumption good. Households can hold capital or one-period bonds (in zero net supply) with interest rate r .

Household consumption choice implies the Euler equation:

$$\beta(1 + r') \left(\frac{C'}{C} \right)^{-\frac{1}{\psi}} = 1. \quad (\text{Euler})$$

We denote r_K the rental rate of capital and r the interest rate on one-period bonds. We introduce a wedge τ between the rental rate of capital and the cash flows that households receive (e.g., a capital income tax). The no-arbitrage condition between capital and bonds yields a valuation equation for the stock of capital:

$$(1 + r')p_K = (1 - \tau)r'_K + (1 - \delta)p'_K. \quad (\text{capital valuation})$$

Households supply specialized labor to both sectors. We denote w_C and w_I the wage in the consumption and investment sectors, and assume that the sectoral labor supply curves are upward-sloping due to comparative advantage. Formally, the representative household chooses a quantity of labor in each sector to maximize labor income $w_C L_C + w_I L_I$ subject to the constraint

$$1 = \left(\pi L_C^{\frac{1+\phi}{\phi}} + (1 - \pi) L_I^{\frac{1+\phi}{\phi}} \right)^{\frac{\phi}{1+\phi}}, \quad \pi \in (0, 1), \phi > 0. \quad (\text{labor supply})$$

The parameter ϕ corresponds to the elasticity of sectoral labor supply. The limit $\phi \downarrow 0$ gives fixed shares: labor cannot get reallocated across sectors. In contrast, the case $\phi \rightarrow \infty$ implies perfect mobility, i.e. $1 = \pi L_C + (1 - \pi) L_I$.

This specification for the sectoral labor supply could be micro-founded via a Roy-Fréchet selection model. Formally, consider a continuum of households that draw a productivity pair (a_C, a_I) in each sector according to a bivariate Fréchet distribution with shape $1 + \phi$. Each household chooses the sector in which they maximize their labor income — $a_C w_C$ if they work in the consumption sector and $a_I w_I$ if they work in the investment sector. Then it is well known that the labor supply side of the economy is equivalent to the one specified in (labor supply) for a representative household (see, for instance, [McFadden, 1974](#), [Eaton and Kortum, 2002](#)).

Equilibrium. An equilibrium is a sequence $\{K_C, K_I, K, L_C, L_I\}$ and prices $\{w_C, w_I, r_K, p_K, r\}$ such that firms minimize costs, households choose consumption and sectoral labor supply, markets clear, and equations (capital accumulation) and (capital valuation) hold. There are three intertemporal equations (i.e. equations that relate variables across different periods): (capital valuation), (capital accumulation), and (Euler). The rest of the conditions constitute the intratemporal block of our model, which pins down allocations and prices at each date given the state of the economy, as summarized by p_K and K .

In the rest of the section, we focus on the long run (i.e., the steady-state of the economy with $C = C'$, $p_K = p'_K$ and $K = K'$).² The intertemporal equations simplify to:

$$r = \frac{1}{\beta} - 1, \quad I = \delta K, \quad p_K = (1 - \tau) \frac{r_K}{r + \delta}.$$

The first equation says that the long-run interest rate is pinned down by the household's preference parameters. The second equation says that the long-run investment-to-capital ratio is pinned down by the depreciation rate of capital. The third equation says that the long-run price of capital is the ratio between its cash-flow and the sum of the interest rate and the depreciation rate. One way to understand this equation is to realize that capital is an intermediary input; like all intermediary inputs, its price is equal to its marginal return in production, with the key subtlety that capital can be used in all future periods, and so its price corresponds to the present value of its future returns.

Comparative statics. We are interested in how capital prices and quantities respond to a long-run shift in the demand for capital, which we capture through an exogenous perturbation $d\tau$ around a steady state with $\tau = 0$. Note that the parameter τ only affects the equilibrium through the capital valuation equation. It corresponds to a *shift in the demand for capital* which leaves the supply side completely unchanged. In terms of our comparative static experiment, $d\tau$ is isomorphic to changes in the representative household's β (e.g., due to demographic change, wealth concentration, financial repression) or to other wedges in the demand for capital.³

In the neoclassical growth model, consumption and investment goods are produced with the same technology, so the supply side pins down the relative price of investment goods. In steady state, (capital valuation) then implies that, for any decrease in capital tax τ , the gross rental rate r_K decreases one-for-one so that the after-tax rental rate $r_K(1 - \tau)$ is unchanged. The capital stock then adjusts so that its marginal product equals this new rental rate. Hence, the long-run elasticity of the capital-good price is zero, and the long-run elasticity of the capital stock is $1/(1 - \alpha)$, the

²In Appendix A, we characterize the general equilibrium sequence and its comparative static saddle paths.

³More precisely, for any endogenous object X such as capital quantity or its price, we have

$$\frac{d \log X}{d \log (\beta^{-1})} = - \frac{1 + r}{r + \delta} \frac{d \log X}{d \log (1 - \tau)}.$$

Both comparative statics are the same up to a scalar.

inverse of one minus the Cobb-Douglas exponent on capital (i.e., the inverse of the labor share).

In a multi-sector model, however, the technology side no longer pins down the relative price of capital. When demand for savings increases, the economy reorients factors of production from the consumption to the investment sector, but that increases the relative cost of producing investment goods relative to consumption goods. As a result, the supply of capital is not perfectly elastic and some of the shock in the demand for capital is absorbed through higher capital prices rather than higher quantities.

The following proposition determines the long-run elasticities of capital prices and quantity in the stylized model. We denote by λ_C and λ_I the steady-state nominal output shares of the two sectors and by λ_K and λ_L the steady-state income shares of capital and labor. We impose the stability condition from Appendix A.1, $\frac{\alpha_C - \alpha_I}{1 - \alpha_I} + \frac{1}{1 + \phi} \frac{1 - \alpha_C}{\alpha_C} \frac{\lambda_K}{\lambda_C} > 0$.

Proposition 1.1 (Long-run elasticities). *The long-run elasticities of the price and quantity of capital with respect to a capital-demand shock are:*

$$\begin{aligned} \frac{d \log p_K}{d \log(1 - \tau)} &= \underbrace{\frac{\alpha_C - \alpha_I}{1 - \alpha_I}}_{\text{technology asymmetry}} + \underbrace{\frac{1}{1 + \phi} \frac{1 - \alpha_C}{\alpha_C} \frac{\lambda_K}{\lambda_C}}_{\text{labor specialization}}, \\ \frac{d \log K}{d \log(1 - \tau)} &= \frac{1}{\lambda_L} \left(1 - \underbrace{(\alpha_C - \alpha_I) \left(\frac{\lambda_C}{1 - \alpha_I} + \frac{\lambda_I}{\alpha_C} \right)}_{\text{technology asymmetry}} - \underbrace{\frac{1}{1 + \phi} \frac{1 - \alpha_C}{\alpha_C} \lambda_K}_{\text{labor specialization}} \right). \end{aligned}$$

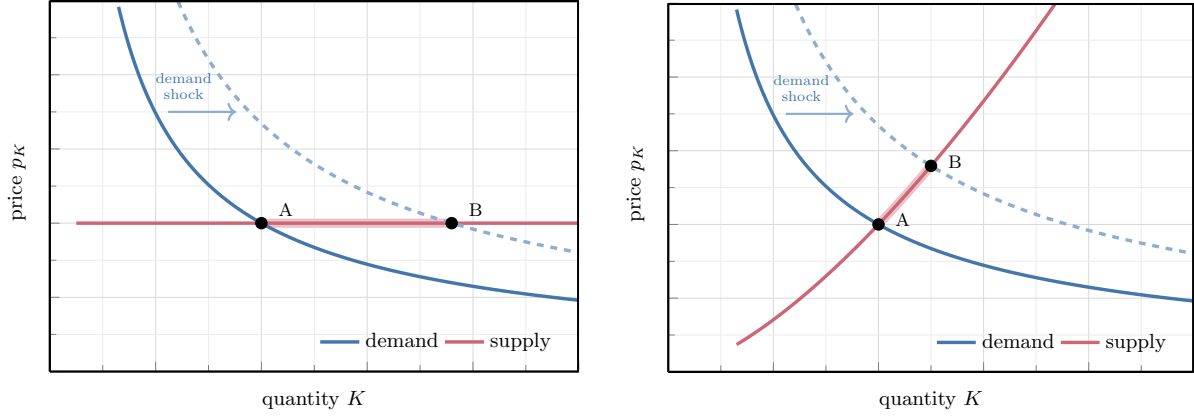
Appendix A.1 derives Proposition 1.1. In the special case of the one-sector neoclassical growth model ($\alpha_C = \alpha_I$ and $\phi = \infty$), the proposition gives the classical result: the elasticity of the capital price is zero, while the elasticity of the capital stock is the inverse of the labor share.

Outside of this special case, the elasticity of the capital price is the sum of two components. The first reflects *technology asymmetry*, and is positive when $\alpha_C > \alpha_I$, i.e., when the investment sector is more labor-intensive than the consumption sector. Intuitively, if this is the case, anything that makes labor more expensive relative to capital (such as a shift in the demand for capital) raises the cost of producing investment goods relative to consumption goods. This is exactly the same logic as in the Stolper-Samuelson theorem in international trade: the price of a good increases with the price of the factor used more intensively in its production.

The second term reflects *labor specialization*, and is positive as long as ϕ is finite. Intuitively, as soon as $\phi < \infty$, it is costly to reallocate labor from C to I , which creates a bottleneck when the economy tries to respond to a shock to the demand for savings by producing more capital. This bids up the price of labor in the investment sector relative to the consumption sector, which in turn pushes up the relative price of capital.

The long-run elasticity of capital quantity is the sum of three terms. The first corresponds to the one-sector neoclassical benchmark, while the remaining two are dampening terms with

exactly the same interpretations as the two components of the price elasticity.⁴



(a) Neoclassical-growth model: $\alpha_C = \alpha_I$ and $\phi = \infty$

(b) Stylized model: $\alpha_C > \alpha_I$ or $\phi < \infty$

Figure 1: Equilibrium in the capital market after a shift in capital demand

The long-run *demand curve* collects the (p_K, K) pairs that satisfy the intratemporal block together with $p_K = (1 - \tau)r_K / (r + \delta)$; the long-run *supply curve* those that satisfy the intratemporal block together with $I = \delta K$ (Appendix A.1).

Figure 1 illustrates the situation graphically, by comparing a shift in the demand for capital in the one-sector neoclassical growth model versus our stylized two-sector model. The more inelastic the capital supply (a steeper supply curve), the more the shock is absorbed through prices rather than quantities. In the neoclassical limit, with perfectly elastic capital supply, demand shocks translate entirely into quantities. In a more realistic model, where the supply of capital is not perfectly elastic, demand shocks are absorbed by a mix of higher quantities and higher capital prices.

Response of factor prices. The positive elasticity of the capital price to demand shocks has direct implications for the incidence of saving (capital-tax) shocks. In the one-sector neoclassical growth benchmark, a decrease in the capital tax is fully absorbed by higher capital quantity (instead of higher capital prices): in the long run, the post-tax rental rate of capital remains the same while the wages of workers increase, reflecting the higher marginal revenue of labor. In our stylized model, however, the supply of capital is less elastic, and so the incidence of lower capital taxes is partly tilted towards capitalists.

To see why, use the fact that, due to unit cost pricing in each sector, the weighted change in

⁴In fact, in this model there is a direct mapping between the elasticity of capital prices and the extent to which the quantity response is dampened:

$$\frac{d \log K}{d \log(1 - \tau)} = \frac{1}{\lambda_L} \left(\frac{\lambda_K}{\alpha_C} - \lambda_C \frac{d \log p_K}{d \log(1 - \tau)} \right).$$

output prices in the economy must equal the weighted change in factor prices:

$$\lambda_I d \log p_K = \lambda_K d \log r_K + \lambda_L d \log w,$$

where $d \log w$ denotes the change in the aggregate wage—the income-share-weighted average of w_C and w_I . Because the consumption good is the numeraire, the left-hand-side only shows the change in the price of the investment good.

Now, in the long run, the steady state valuation equation implies

$$d \log p_K = d \log r_K + d \log(1 - \tau).$$

We can therefore write the change in factor prices as a function of the change in the price of capital:

$$d \log w = \frac{\lambda_K}{\lambda_L} d \log(1 - \tau) - \frac{\lambda_K - \lambda_I}{\lambda_L} d \log p_K.$$

In the neoclassical benchmark, $d \log p_K = 0$, so a decrease in taxes does not affect the post-tax compensation of capital holders while wages increase by λ_K/λ_L . Put differently, workers entirely capture the effect of lower taxes through higher wages. However, in our stylized model with $\alpha_C \neq \alpha_I$ and/or $\phi < \infty$, part of the shock is absorbed by p_K , so the pre-tax rental rate falls by *less than* one-for-one and wages rise by *less than* λ_K/λ_L . In incidence terms, some of the shock is borne by capital holders (through higher post-tax capital payments) rather than workers.⁵ We will quantify the welfare incidence of capital taxes more precisely in the calibrated model in Section 3.

1.2 Extensions

Our stylized model departs from the neoclassical growth benchmark along two axes: (i) heterogeneous factor intensities across the consumption and investment sectors; and (ii) limited sectoral labor mobility. In Appendix A, we augment the stylized model with a number of extensions one by one. We describe below how each extension affects our comparative static of interest.

CES production. The stylized model assumes Cobb-Douglas production. Appendix A.2 derives the linearized equilibrium conditions and long-run elasticities when sectoral production functions are CES. The new sufficient statistics are the sectoral substitution elasticities. When substitution between capital and labor is easier, the quantity response of K is larger, as in the neoclassical benchmark. Nevertheless, factor-intensity heterogeneity ($\alpha_C \neq \alpha_I$) or labor specialization ($\phi < \infty$) continues to dampen the adjustment of K to demand shocks as in the Cobb-Douglas case.

⁵More precisely, the change in the net income earned by a capital holder who maintains their stock of capital constant is

$$d \log ((1 - \tau)r_K) - \frac{\lambda_I}{\lambda_K} d \log p_K = \frac{\lambda_K - \lambda_I}{\lambda_K} d \log p_K.$$

Multiple factors of production. We extend the model with $f \in \{1, \dots, F\}$ fixed factors L_f (e.g., labor by type, land) and $n \in \{1, \dots, N\}$ reproducible factors K_n (e.g., structures, equipment, intangibles). This allows the model to account for sectoral differences in the labor mix (e.g., by education or occupation) and in the composition of capital goods. The key insight for us—which we use in the calibration strategy of Section 3—is that even if total labor shares coincide across sectors C and I , differences in the type of labor hired can make the labor supply faced by a sector less elastic for a given sectoral-equivalent elasticity (e.g., the production of R&D disproportionately requires scientists).

Dynamic multi-occupation labor supply. The canonical Roy-Fréchet labor supply is perhaps better suited for the type of long-run experiment we did thus far, rather than for transitions. Appendix A.4 derives a dynamic occupation block that combines occupation mobility and transition speed in one system. Financial discounting remains governed by r , while the occupation-specific payoff from the current choice decays at rate μ because workers can reoptimize. The transition parameter μ makes labor supply less elastic in the short run without changing the long-run occupation-supply equations. We use this setup in the calibration strategy of Section 3.

Internal adjustment costs and irreversibility. In the stylized model, households own the capital stock and rent it to firms. In practice, firms own their own capital and face internal resource bottlenecks when trying to scale too fast (engineering time, management attention), which drives a wedge between the value of businesses and the replacement cost of their capital stock. Appendix A.5 extends the model with (i) an internal adjustment cost as in the standard q -theory of investment and (ii) irreversible sectoral investments. The calibration strategy of Section 3 uses external evidence on internal adjustment costs to discipline this force, which is separate from our theory which is based on aggregate resource bottlenecks.

Imported inputs and modified Euler equations. Appendix A.6 adds imported inputs as an Armington CES bundle, which makes trade openness a force that can absorb part of the capital-demand shock through foreign goods prices. Appendix A.7 combines the spender-saver aggregation block and the discounted Euler equation. These extensions affect the transition and the mapping from capital-demand shocks to real rates, but the long-run supply elasticities continue to be disciplined by the same production and labor-supply objects.

2 Mapping the theory to the data

Section 1 shows that the long-run supply elasticity of capital depends on two empirical objects: the technology asymmetry between consumption and investment, and the labor specialization that makes reallocation across final uses costly. This section describes how we measure both objects in

U.S. data. The main facts are that investment is more labor-intensive than consumption, and that intangible investment relies disproportionately on specialized occupation bundles.

2.1 Datasets

BEA input–output accounts. We begin with the BEA Use and Make Tables, available annually from 1963 to 2023 (U.S. Bureau of Economic Analysis, 2024a). The Use Table reports intermediate commodity purchases by industries and final uses of each commodity. We group private final uses into consumption and four investment categories: equipment, structures, intellectual property products (IPP), and brand/organizational services. The Make Table maps commodities to the industries that produce them. For the main results, we work at the 71 BEA “summary commodity” level, which is available over the full sample. Appendix B.2 reports robustness using the 402-commodity detail table in 2017.

Following Corrado et al. (2009), we use a broader definition of intangible investment than the BEA. In particular, we reclassify firm expenditures on organizational capital—expenditures related to workforce human capital, distribution systems, logistics, product design, and customer/brand capital—from intermediate expenditures to investment (Atkeson and Kehoe, 2005; Eisfeldt and Papanikolaou, 2013; Crouzet and Eberly, 2021; Crouzet et al., 2022). Concretely, we focus on two commodities purchased by industries: “Miscellaneous professional, scientific, and technical services” and “Management of companies and enterprises.” The first category includes externally purchased services such as design services, management consulting, advertising, and computer system design. The second captures the service output of establishments that manage other units within a company.

KLEMS. We merge the input–output accounts with the Integrated Industry-Level Production Account, available annually from 1963 to 2023 (U.S. Bureau of Economic Analysis, 2024b). KLEMS allocates each industry’s revenues into payments to high school workers, college-educated workers, capital income, and intermediate expenditures. We split the college category into college and post-college workers using CPS wage records. We also split KLEMS capital income into land, equipment, structures, IPP, and organizational capital. Land payments are set to 18% of revenues in agriculture, 24% in residential housing, and zero elsewhere. The remaining capital income is allocated across reproducible capital types in proportion to scaled industry investment rates, using BEA Fixed Asset Tables for equipment, structures, and IPP, and the input–output accounts for organizational services.⁶

⁶In steady state, capital income equals the investment rate multiplied by $(r + \delta) / (g + \delta)$, where r is the interest rate, δ the depreciation rate, and g the growth rate. We therefore split capital income shares using these scaled investment rates for each capital type.

CPS. We use CPS ASEC worker records to measure labor heterogeneity and occupation mobility. For industry labor shares, CPS wage records from 1968 to 2023 split labor compensation into high school, college, and post-college groups; for 1963–1967, we use the 1968 CPS split so that the BEA/KLEMS panel remains balanced. For labor specialization, the key CPS feature is that ASEC reports both current occupation and previous-year occupation. We use workers ages 25–54 with valid education, weights, and current/previous-year occupations, and pool 1976–2023 to construct occupation transition matrices. We map occupations to the 67 occupation groups of [Hsieh et al. \(2019\)](#) and drop the home group, leaving the 66 market occupations used throughout; we aggregate to the HHJK broad groups for paper figures. Sector occupation bundles use occupation-payroll shares by BEA summary industry, education, and occupation; the CPS transition graph then says which occupation differences are easy or hard to bridge.

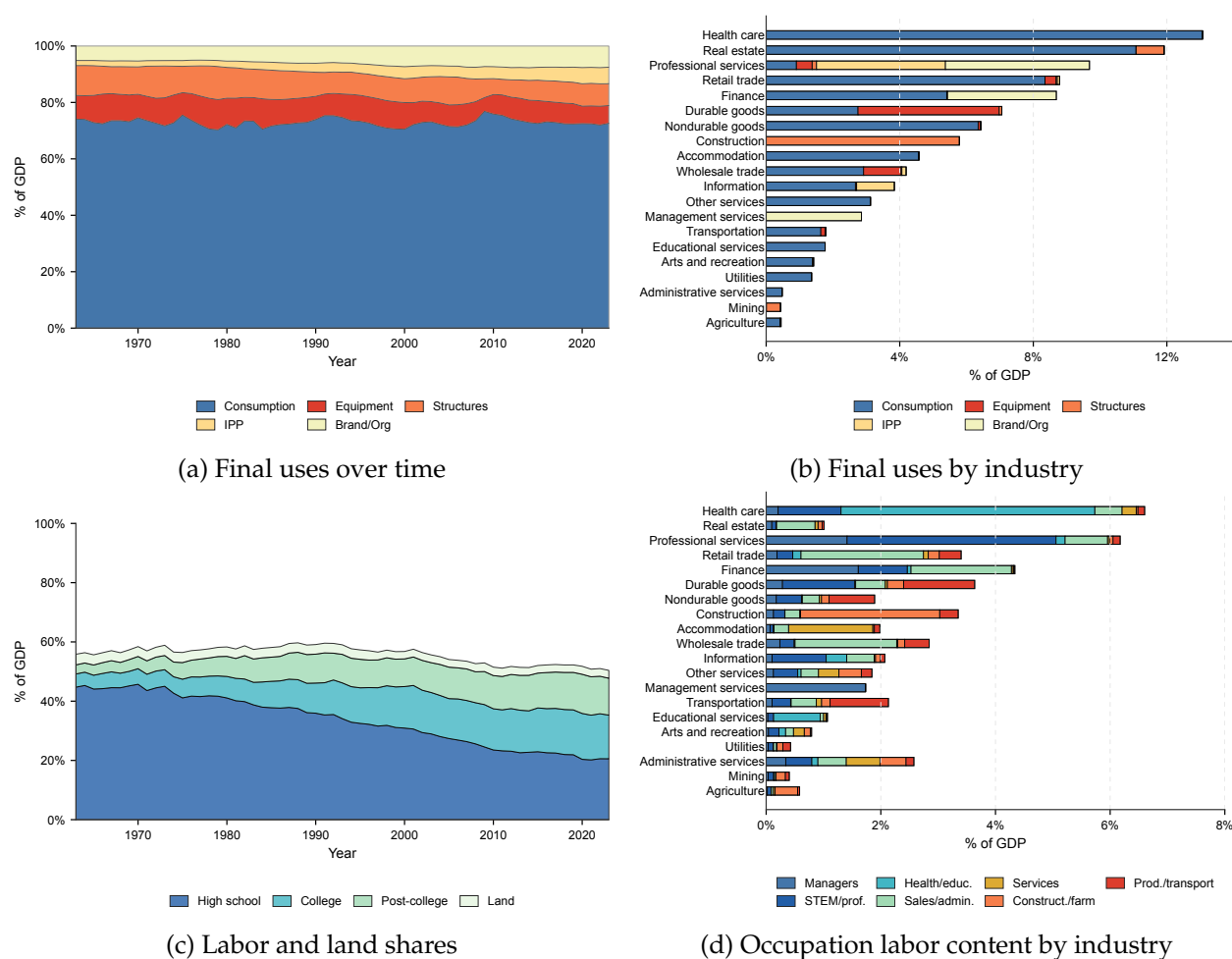


Figure 2: Composition of U.S. value added and labor inputs

Figure 2 maps the raw data into the objects used below. Subfigure 2a plots private final uses over time. Total investment, given by the non-consumption components, averages about 20% of GDP and is fairly stable over time; equipment and structures have declined secularly, while IPP

and organizational services have risen. Subfigure 2b shows that structures are concentrated in construction, equipment in durable goods, and IPP/organizational services in information and professional services. Subfigure 2c plots labor by education group and land as shares of GDP, leaving the capital residual unshaded. Subfigure 2d reports occupation-family labor content by industry, also as a share of GDP.

2.2 Methodology

Factor shares. We compute factor shares for consumption and investment in three steps. First, we convert industry factor shares from KLEMS into commodity factor shares. Since industries can produce multiple commodities, we use the BEA Make Table and adopt the common commodity-technology assumption: each commodity has a unique input structure, regardless of which industry produces it. Appendix B.1 gives the matrix details and reports robustness to the common industry-technology assumption.

Second, we convert direct commodity factor shares into *total-requirements* factor shares, which include both direct factor use and the factors embodied in intermediate inputs. We do so by multiplying direct-requirements factor shares by the Leontief inverse of the commodity-by-commodity requirement matrix. Figure 3 shows this adjustment directly, comparing factor shares before and after consolidating intermediate inputs.

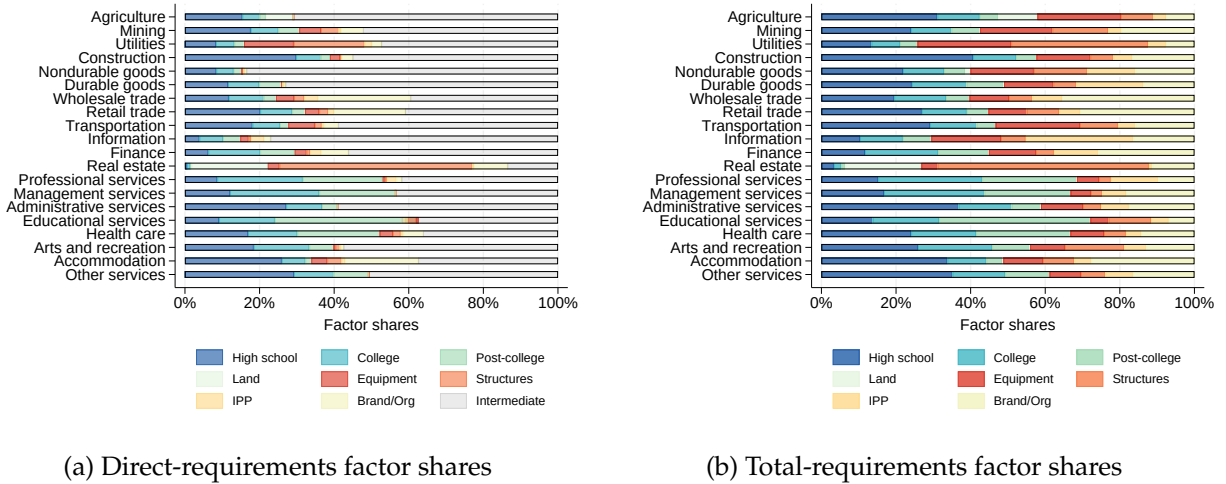


Figure 3: Direct and total factor requirements by industry

Third, we aggregate commodity factor shares to final uses using each commodity's share in the relevant final-use bundle. This produces the sector factor shares α used in the theory: consumption, aggregate investment, and the four investment types.

We extend the same factor-share accounting within labor. We split each sector-by-education labor share into occupation cells and apply the same total-requirements calculation to obtain occupation bundles γ_{sfo} for consumption and each investment type. For each education group,

we measure occupational dissimilarity using the standard index of [Duncan and Duncan \(1955\)](#), applied to consumption and investment type J :

$$\frac{1}{2} \sum_o |\gamma_{Jfo} - \gamma_{Cfo}| = 1 - \sum_o \min\{\gamma_{Jfo}, \gamma_{Cfo}\}.$$

The measure is zero when the two occupation bundles coincide and one when they have no occupation overlap. Equivalently, a value of 0.4 means that 40% of either normalized occupation bundle cannot be matched occupation-for-occupation with the other. We average the education-specific values using each group’s share of the combined consumption and investment-type wage bill. This is a descriptive factor-share comparison: it uses no transition data and requires no mapping from CPS flows into labor-supply elasticities.

The quantitative model subsequently combines these occupation bundles with CPS occupation transitions to discipline which occupation differences are easy or hard to bridge. [Appendix B.1.2](#) describes that separate mapping into occupation cross-elasticities.

2.3 Evidence on technology asymmetries and labor specialization

[Table 1](#) reports the resulting factor shares. The first row reports factor shares for the aggregate private economy, defined as a final-use-weighted average across commodities used in private consumption or private investment. We find an average labor share of 49%, which is somewhat lower than common benchmark numbers (see, e.g., [Koh et al. \(2020\)](#))—mainly because, relative to much of the literature, we include housing and utilities, which have relatively low labor shares. The next two rows report factor shares for the consumption and investment sectors. The labor share is 45% in consumption versus 59% in investment, a difference of 14 pp. Most of this gap reflects differences in skilled labor: the college labor share is 11 pp higher in investment than in consumption.

The remainder of [Table 1](#) reports factor shares for equipment, structures, IPP, and brand/organizational investment. The higher labor share of investment is driven almost entirely by intangible investment: the average labor share for tangible investment (equipment + structures) is 52%, while it rises to 67% for intangible investment (IPP + organization). The capital columns show that consumption and investment have similar intensities for equipment and organizational capital. Consumption relies more on structures, reflecting housing consumption, while investment—through IPP-producing sectors—relies more on IPP capital.

[Figure 4](#) shows which broad occupations underlie the factor-share differences. Investment is tilted toward executives, construction, engineering, and management-related occupations, while consumption is tilted toward health, services, sales, and education. The bars report occupation labor content by final use as a share of GDP, so the occupation comparison uses the same total-requirements accounting as [Table 1](#).

[Figure 5](#) summarizes the two empirical objects that map into the theory. The left panel plots

Table 1: Factor shares by sector (2014–2023)

	C+I share	Factor shares							
		Labor			Land	Capital			
		HS	College	Post-college		Equipment	Structures	IPP	Brand/Org
$C + I$	1.00	0.22	0.15	0.12	0.03	0.12	0.15	0.09	0.12
C	0.73	0.21	0.13	0.12	0.04	0.12	0.19	0.07	0.12
I	0.27	0.24	0.20	0.15	0.00	0.12	0.06	0.12	0.12
$I_{\text{Equipment}}$	0.07	0.23	0.16	0.10	0.00	0.14	0.07	0.16	0.14
$I_{\text{Structures}}$	0.08	0.37	0.12	0.07	0.00	0.16	0.09	0.05	0.14
I_{IPP}	0.05	0.15	0.26	0.21	0.00	0.08	0.04	0.17	0.09
$I_{\text{Brand/Org}}$	0.08	0.17	0.28	0.24	0.00	0.07	0.04	0.11	0.09

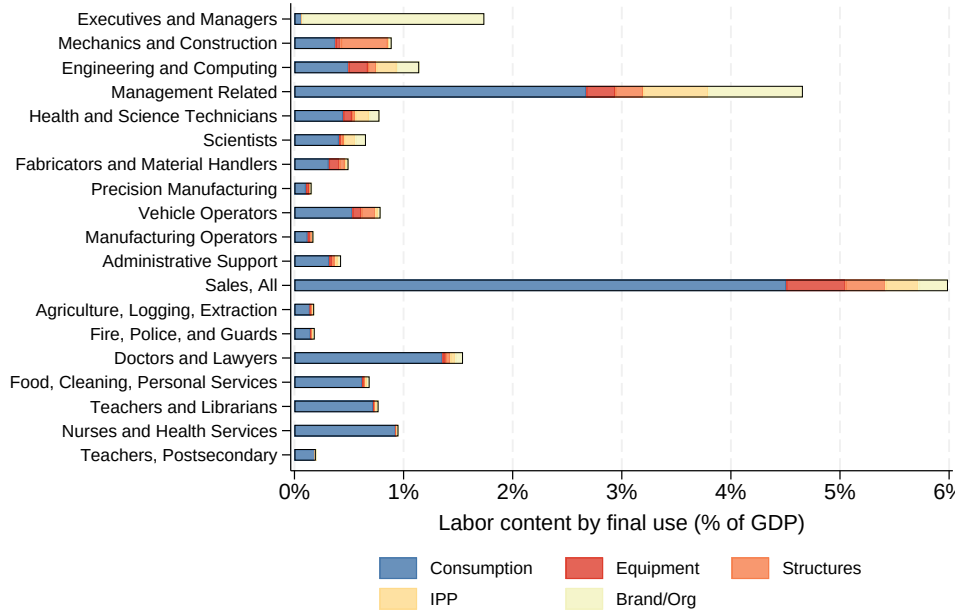


Figure 4: Occupation labor content by final use (2014–2023)

Notes: Occupations are sorted by the ratio of aggregate investment to consumption occupation shares.

each investment type's labor share minus the consumption labor share. The black line aggregates investment types using their final-use shares and corresponds to $\alpha_C - \alpha_I$ in the stylized two-factor model; it rises from 7 pp at the start of the sample (1963–1968) to 14 pp at the end (2014–2023). The right panel reports occupational dissimilarity between consumption and the same investment types, along with aggregate investment in black.

The two panels deliver a direct descriptive analogue of the model's forces. Investment is more labor-intensive than consumption, and producing each investment type requires an occupation bundle that differs materially from consumption's. Brand and organizational investment is the most dissimilar, while equipment is the least. These comparisons establish technology asymmetry and labor specialization without identifying a sectoral labor-supply elasticity. The quantitative

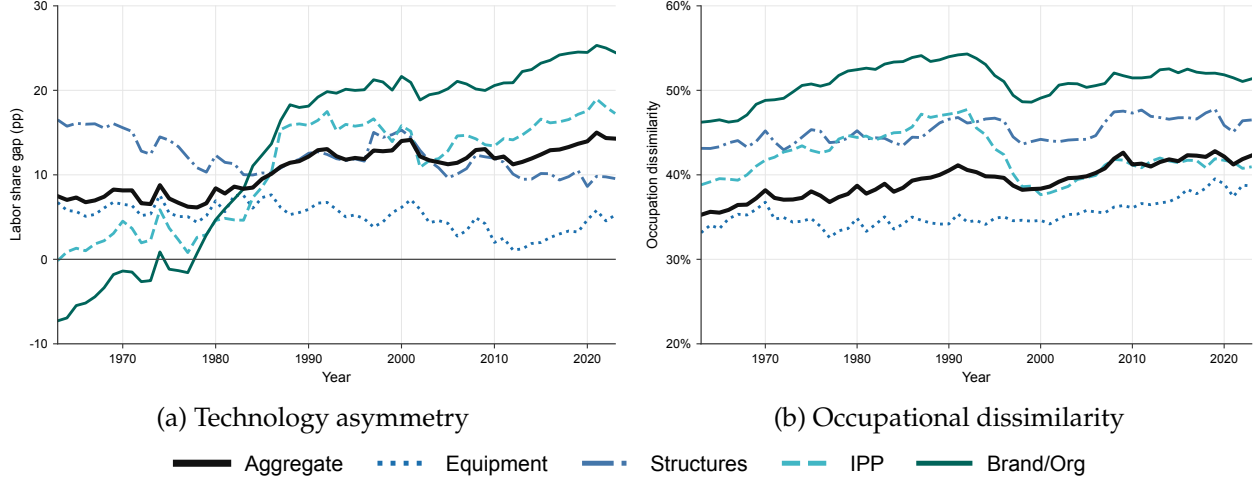


Figure 5: Technology asymmetry and occupational dissimilarity

Notes: The left panel is in percentage points. In the right panel, the black line first forms aggregate investment's occupation bundle as the wage-bill-weighted average of the four investment-type bundles, then compares it with consumption.

model adds that structural step by combining the same occupation bundles with the CPS transition graph.

Appendix B.2 reports robustness checks. We consider not capitalizing organizational services, capitalizing a broader set of financial services, collapsing private and public sectors, replacing the common commodity-technology assumption with a common industry-technology assumption, using the detailed 402-commodity table, and treating imports as a separate input. Across these alternatives, investment remains more labor-intensive than consumption; the main exception is that if organizational services are not capitalized, the rise in the investment-consumption labor-share gap is much smaller.

3 Quantitative model

We now calibrate an extended version of the stylized model from Section 1, quantify the implied dynamic response to demand shocks, and finally simulate the welfare incidence of capital taxation.

We introduce $N = 4$ capital types (equipment, structures, IPP, organizational), $1 + N = 5$ sectors (consumption and four investment goods), and three education groups. The labor block has heterogeneity in factors: demand shocks for different goods affect different workers (a demand shock for intangible capital disproportionately benefits scientists, a housing shock benefits construction workers). The data on labor mobility, however, lives at the occupation level rather than the sector level. We therefore take occupations as the model's primitive labor margin, and report sectoral-equivalent elasticities only as interpretive summaries.

Concretely, production uses factor cells L_{sf_o} , where $s \in \{C, I_1, \dots, I_N\}$ is the sector, f is education, and o is occupation. The factor share of cell (f, o) in sector s is $\alpha_{sL_{f_o}} = \alpha_{sL_f} \gamma_{sf_o}$. Worker cells are mobile across sectors within an occupation cell, so they earn a common return w_{f_o} ; sectoral oc-

cupation demands aggregate to the occupation stock L_{fo} . Worker reallocation across occupations is governed by education-specific cross-elasticities $\phi_{foo'}$. The full quantitative model expands each education group into $O = 66$ occupation cells and appends land as a singleton factor cell h with zero labor-supply elasticities ($\eta_h = \mu_h = 0$) and zero cross-elasticities; throughout the equations, Σ_{fo} includes this land cell and w_h denotes its rent.

Table 2: Linearized model equations

Intratemporal block	Equation
C pricing	$0 = \sum_m \alpha_{CK_m} d \log r_{K_{Cm}} + \sum_{fo} \alpha_{CL_{fo}} d \log w_{fo}$
I_n pricing	$d \log p_{I_n} = \sum_m \alpha_{I_n K_m} d \log r_{K_{I_n m}} + \sum_{fo} \alpha_{I_n L_{fo}} d \log w_{fo}$
K_{sn} demand	$d \log K_{sn} = d \log Y_s + \sum_{m \neq n} \alpha_{s K_m} d \log \frac{r_{K_{sm}}}{r_{K_{sn}}} + \sum_{fo} \alpha_{s L_{fo}} d \log \frac{w_{fo}}{r_{K_{sn}}}$
L_{sfo} demand	$d \log L_{sfo} = d \log Y_s + \sum_m \alpha_{s K_m} d \log \frac{r_{K_{sm}}}{w_{fo}} + \sum_{f'o' \neq fo} \alpha_{s L_{f'o'}} d \log \frac{w_{f'o'}}{w_{fo}}$
L_{fo} clearing	$d \log L_{fo} = \sum_s \frac{\lambda_s \alpha_{s L_{fo}}}{\lambda_{L_{fo}}} d \log L_{sfo}$
Intertemporal block	Equation
K_{sn} valuation	$d \log((1 + r_{t+1})p_{K_{sn},t}) = \frac{\lambda_{K_{sn}}}{\lambda_{W_{sn}}} d \log((1 - \tau_{n,t+1})r_{K_{sn},t+1}) - \frac{\lambda_{I_{sn}}}{\lambda_{W_{sn}}} d \log p_{I_{n,t+1}} + \frac{1}{1+r} d \log p_{K_{sn},t+1}$
K_{sn} accumulation	$d \log K_{sn,t+1} = (1 - (1 - \theta_n)\delta_n) d \log K_{sn,t} + (1 - \theta_n)\delta_n d \log I_{sn,t}$
I_{sn} investment-q	$d \log I_{sn,t} = d \log K_{sn,t} + \frac{1}{\theta_n} d \log \frac{p_{K_{sn},t}}{p_{I_{n,t}}}$
C euler equation	$d \log C_t = d \log C_{t+1} - \psi d \log(1 + r_{t+1})$
L_f accumulation	$d \log L_{f,t+1} = (1 - \nu) d \log L_{f,t} + \nu \eta d \log \frac{p_{L_{f,t}}}{p_{L,t}}$
L_{fo} accumulation	$d \log L_{fo,t+1} = d \log L_{f,t+1} + (1 - \mu_f) d \log \frac{L_{fo,t}}{L_{f,t}} + \mu_f \sum_{o'} \phi_{foo'} d \log \frac{p_{L_{fo},t}}{p_{L_{f,t}}}$
L_{fo} valuation	$d \log p_{L_{fo},t} = \frac{r + \mu_f}{1+r} d \log w_{fo,t+1} - d \log(1 + r_{t+1}) + \frac{1 - \mu_f}{1+r} d \log p_{L_{fo},t+1}$

$d \log x$ denotes a log-deviation from steady state, Y_s is sector s 's output (C or I_n), and λ denotes an income share. The sector-type shares $\lambda_{W_{sn}}$, $\lambda_{I_{sn}}$, and $\lambda_{K_{sn}}$ are defined in Appendix A.5. The value indexes are $d \log p_{L_{f,t}} = \sum_o (\lambda_{L_{fo}} / \lambda_{L_f}) d \log p_{L_{fo},t}$ and $d \log p_{L,t} = \sum_f [\lambda_{L_f} / \sum_{f'} \lambda_{L_{f'}}] d \log p_{L_{f,t}}$.

To simulate transition dynamics, we augment the model with a dynamic labor supply block where cohorts sort into education groups and then across occupations in response to relative wages. Labor specialization is induced by sectors' different occupation bundles. We also include sector-specific capital stocks subject to internal adjustment costs, aimed at matching firm-level evidence on sluggish investment responses to Tobin's q .

Linearized equations. We focus directly on the linearized model around the steady-state in response to an arbitrary sequence of type-specific demand shocks $\{\{d \log(1 - \tau_{n,t})\}_{n=1}^N\}_{t=0}^\infty$. As in the stylized model, this shock maps directly to a capital income tax shock, but is also isomorphic to Euler equation shocks. Since the model is a direct extension of the stylized model from Section 1, we list the linearized model equations in Table 2, and refer the reader to Appendix A.3, Appendix A.4, and Appendix A.5 for formal derivations of the extensions. The new parameters of the labor block will be introduced formally in the calibration section below.

The equations are broken down into the intratemporal block (which determines the allocation given the state of the economy) and the intertemporal block (which pins down the evolution of the state). The states are the stocks of capital and labor, and the co-states are the associated present values of future income flows from these factors. Table 2 keeps the production side compact; the quantitative system expands the labor equations to the occupation cells described above.

3.1 Calibration

First, we calibrate the demand side of the model: a real annual return of $r = 0.03$ (the 2017–2023 average of the time-series measure of $r - g$ constructed in Appendix C.4) and a unit intertemporal elasticity of $\psi = 1$. The goal is to first isolate the unique predictions of the supply side of the model (i.e., how does production of consumption and investment goods respond to relative prices). Then, we consider alternatives to the representative-agent Euler equation using the joint spender-saver discounted-Euler block of Appendix A.7: hand-to-mouth consumption raises the aggregate marginal propensity to consume, while discounting the Euler equation attenuates the effect of distant real rates. The validation exercises in Appendix C.3 retain this intertemporal specification in the model column.

Second, we calibrate the sectoral factor shares and output shares using the BEA data described in Section 2, averaged over the 2017–2023 period. We do the same for the depreciation rates from the BEA over the 2017–2023 period, which average 14% for equipment, 3% for structures, and 24% for IPP. For organization capital, we impute the same depreciation rate as IPP. (Table 1 displays factor shares for the longer 2014–2023 window; the occupation-level labor moments described below are also measured over 2014–2023.)

Table 3 reports the calibration of the remaining parameters (i.e., the labor supply matrices, transition speeds, and the capital adjustment parameter). Below, we go over the existing evidence in detail and describe the moments that we seek to match.

Table 3: Baseline calibration and empirical targets

Parameter	Value	Targeted moment	Reference
α, λ, δ	Table 1	sectoral factor/output shares	BEA-KLEMS
ν	0.03	cohort replacement rate of 3% annual	Adão et al. (2024)
η	0.4	education choice elasticity of 0.4	Adão et al. (2024)
μ_f	(0.08, 0.07, 0.06)	annual occupation transition speeds	CPS
$\phi_{f_{oo'}}$	Table A1	occ. cross-elasticities	CPS
θ_n	(0.17, 0.04, 0.27, 0.27)	quadratic adjustment cost of 1.5	Chodorow-Reich (2026)

Appendix B.1.2 describes the CPS construction and the projection to the sectoral-equivalent elasticities ϕ_f reported below.

Calibration of the dynamic labor block. The dynamic labor block has four primitive parameters: a cohort replacement rate ν , an education-choice elasticity η , education-specific occupation transition speeds μ_f , and education-specific occupation cross-elasticities $\phi_{f00'}$. Workers respond to wage changes along two margins, education and occupation; (ν, μ_f) set the speeds and $(\eta, \phi_{f00'})$ set the long-run elasticities. In the occupation valuation equation, the occupation-specific payoff from today's choice survives only until the next reoptimization opportunity, so the continuation value of that payoff is multiplied by $1 - \mu_f$. The timing is as follows: each period, a fraction ν of each education group's workforce is replaced by entrants, who sort across education groups with elasticity η ; independently, workers in education group f reoptimize their occupation with probability μ_f , with long-run cross-elasticities $\phi_{f00'}$.

The long-run occupation elasticities $\phi_{f00'}$, together with sectors' occupation bundles γ_{sfo} , effectively generate sectoral labor supply elasticities. To build intuition, project the occupation block onto a single consumption-versus-investment margin within each education group. Under the projection, each education group f has a scalar long-run sectoral elasticity ϕ_f , and the partial-equilibrium response of sector-by-education labor to a permanent wage change separates additively into an education and a sectoral margin:

$$d \log L_{sf,t} = \underbrace{\left(1 - (1 - \nu)^t\right) \eta d \log \left(\frac{w_f}{w}\right)}_{\text{education choice}} + \underbrace{\left(1 - (1 - \mu_f)^t\right) \phi_f d \log \left(\frac{w_{sf}}{w_f}\right)}_{\text{sectoral-equivalent labor supply}},$$

where w_{sf} is the projected sector-by-education wage index, w_f is its factor-share-weighted average across sectors, and w is the aggregate labor wage. Each term approaches its long-run value geometrically at speed ν and μ_f , respectively.

We set $\nu = 0.03$ and $\eta = 0.4$ following [Adão et al. \(2024, Table 2\)](#). The occupation transition speeds are calibrated from the model-consistent steady-state identity applied to CPS one-year occupation transitions ([Appendix B.1.2](#)), giving $\mu_f = (0.08, 0.07, 0.06)$ for high school, college, and post-college workers. The cross-elasticities $\phi_{f00'}$ are constructed from the same CPS transition matrices using the pairwise-flow methodology of [Böhm et al. \(2025\)](#). Combined with the sectoral occupation shares γ_{sfo} , they imply aggregate investment-versus-consumption sectoral-equivalent elasticities $\phi_f = (6.47, 6.04, 3.22)$ for high school, college, and post-college workers. These scalars summarize the occupation block for interpretation; the quantitative model solves the full occupation block directly.

Calibration of adjustment cost parameter. We calibrate θ to match external evidence on capital adjustment costs. Let χ denote the quadratic adjustment-cost coefficient, so the slope of the investment rate $p_I I/K$ on Tobin's q is $1/\chi$. The investment- q equation shows that $1/\theta = (1/\chi) \cdot (p_K K)/(p_I I)$, so $1/\chi$ is the empirically observed counterpart of $1/\theta$ rescaled by the capital-to-investment ratio. [Chodorow-Reich \(2026\)](#) synthesizes the empirical literature; the cleanest IV

estimate, from [Cummins et al. \(1996\)](#) who use tax reforms as instruments, finds an investment- q slope of 0.65, implying $\chi \approx 1/0.65 \approx 1.5$. Our installation function is equivalent to a quadratic adjustment-cost model ([Appendix A.5](#)), so we solve for type-specific θ_n from the steady-state mapping $\theta_n = \delta_n \chi / (1 + \delta_n \chi)$ ([Appendix C.4](#)). Given our BEA depreciation rates (averaged over 2017–2023), this gives $\theta \approx 0.17$ for equipment, 0.04 for structures, and 0.27 for IPP and organizational capital.⁷

3.2 How elastic is the supply of capital?

We start by quantifying the response of capital and valuations to a permanent demand shock $d \log(1 - \tau) = 1_{t \geq 0}$ and focus on aggregate quantities and prices.⁸ Throughout this section, we will refer to [Table 4](#), which summarizes the term structure of these responses in the calibrated model (e.g., on impact, at a planning horizon of 10 years, and in the long run). To tease out which of the two forces emphasized in [Section 1](#) matters more quantitatively, we calibrate two alternative models where we add one ingredient at a time to the neoclassical growth model. Similarly, we consider a model extension with only capital adjustment costs (e.g., Hayashi’s q -theory of investment).

The sensitivity panels of [Table 4](#) report alternative modified Euler equations, trade openness, production substitution, and labor mobility. First, we vary the elasticity of intertemporal substitution $\psi \in \{0.5, 2\}$, bracketing the baseline $\psi = 1$ (log utility). Second, we use the joint spender-saver discounted-Euler block of [Appendix A.7](#). We report spender-saver rows with hand-to-mouth agents consuming an ω share of fixed-factor income, including land, while savers own reproducible capital and follow the Euler equation.⁹ We also report modified Euler equations with $\rho \in \{0.98, 0.95\}$, which attenuate the effect of distant real rates and can move the long-run real rate. The supply-side extensions introduce imports as a CES composite with domestic factors, with Armington trade elasticity $\sigma_M \in \{3, 5\}$ and sector-specific import shares from BEA domestic input–output tables (2017–2023 average; see [Table A5](#)). They also scale the occupation cross-elasticities by two and by one half, which changes the long-run capital-price response by -2 and +2 percentage points, respectively.

Overall, the calibrated model predicts a weak response of capital accumulation, even in the long run (i.e., 1.51 versus 1.94 in the neoclassical growth model). Over a more realistic planning horizon of ten years (i.e., the medium run), the gap between the calibrated model and the neoclassical growth model is even more severe (0.73 versus 1.53). What this means is that back-of-the-envelope calculations regarding the predicted effect of a policy on capital accumulation are

⁷ Adjustment costs are larger for faster-depreciating capital because a given friction χ requires higher θ when δ is larger. The resulting heterogeneity is consistent with evidence that intangible capital has higher adjustment costs ([Belo et al., 2022](#); [Crouzet and Eberly, 2023](#)).

⁸ We define $d \log K_t \equiv \sum_n \frac{\lambda_{K_n}}{\lambda_K} d \log K_{n,t}$ and $d \log p_{K,t} \equiv \sum_n \frac{\lambda_{K_n}}{\lambda_K} d \log p_{K_{n,t}}$ the aggregate quantity and price responses, where the weights are steady-state capital income shares.

⁹ We report $\omega = 0.25$ (baseline, from [Kaplan and Violante \(2014\)](#)) and $\omega = 0.50$ (robustness, approximately matching the annual impact aggregate MPC of [Fagereng et al. \(2021\)](#)). The representative-agent case is nested when $\omega = 0$.

Table 4: Summary of price and quantity responses (permanent demand shock)

	On impact	Medium-run		Long-run	
	d log $p_{K,0}$	d log $p_{K,10}$	d log K_{10}	d log $p_{K,\infty}$	d log K_{∞}
Calibrated model	1.12	0.38	0.73	0.30	1.51
Neoclassical growth model	0.00	0.00	1.53	0.00	1.94
+capital adjustment costs	0.27	0.07	1.42	0.00	1.94
+technology asymmetry	0.02	0.16	1.37	0.19	1.66
+labor specialization	0.87	0.24	1.07	0.11	1.84
Demand-side sensitivities					
intertemporal elasticity $\psi = 0.5$	+3	-6	-8	0	0
intertemporal elasticity $\psi = 2$	-3	+4	+5	0	0
spender-saver $\omega = 0.25$	+4	0	0	0	0
spender-saver $\omega = 0.50$	+9	-1	0	0	0
discount factor $\rho = 0.98$	-1	+1	0	+3	+12
discount factor $\rho = 0.95$	-4	+2	+1	+10	+35
Supply-side extensions					
imported inputs $\sigma_M = 3$	+3	-2	+11	-2	+30
imported inputs $\sigma_M = 5$	+9	-1	+19	-2	+60
capital-labor elasticity $\sigma_{KL} = 0.5$	-18	-10	-24	-5	-72
capital-labor elasticity $\sigma_{KL} = 1.5$	+11	+7	+16	+5	+67
occupation transition speeds $2 \times \mu_f$	-1	-2	+1	0	0
occupation transition speeds $10 \times \mu_f$	-2	-3	+3	0	0
occupation elasticities $2 \times \phi_{f00'}$	-1	-3	+1	-2	+3
occupation elasticities $0.5 \times \phi_{f00'}$	+1	+2	-1	+2	-2
Alternative data construction					
no capitalization of org. services	-11	-6	-4	-10	+2
capitalization of financial services	+6	+1	+3	0	+6
consolidate government sector	-1	-3	+1	-5	+1
common industry-tech. assumption	+1	-2	+1	-5	+7

The first panel reports levels: the fraction of the demand shock absorbed by prices, and log-point changes for quantities. The remaining panels report signed deviations from the calibrated baseline in percentage points ($100 \times$ the difference).

materially lower in the calibrated model relative to the benchmark.

This weak response of real quantities implies a large response of relative prices: a significant part of the demand shock is capitalized in higher capital goods prices. Recall that, in the neoclassical growth model, there is no change in the relative price of capital goods versus consumption goods. In the calibrated model, a fraction 0.30 of the demand shock is absorbed by higher prices in the long run. What this means is that, even after the economy has fully adjusted to the demand shock, capital goods prices remain sensitive to demand conditions. In the short run capital goods owners experience a large revaluation gain of 1.12, which decays toward its long-run level over roughly a decade (i.e., 0.38 after ten years).

Decomposition of the long-run using sufficient statistic. We now decompose the long-run price effect into its fundamental drivers. Conditional on the steady-state valuation shares $\lambda_{K_{sn}}$, $\lambda_{I_{sn}}$, and

$\lambda_{W_{sn}}$, the long run is invariant to capital adjustment costs θ and intertemporal substitution ψ . Recomputing those shares after changing θ instead changes the steady-state linearization point. Holding that point fixed lets us decompose the long-run price effect into technology asymmetry ($\alpha_C > \alpha_I$), labor specialization, and capital-type heterogeneity.

$$\frac{d \log p_K}{d \log(1 - \tau)} = \underbrace{\frac{\alpha_{CK} - \alpha_{IK}}{1 - \alpha_{IK}}}_{=0.18} + \underbrace{\frac{1}{1 + \sum_f \frac{\lambda_{Lf}}{\lambda_L} \phi_f} \frac{1 - \alpha_{CK}}{\alpha_{CK}} \frac{\lambda_K}{\lambda_C}}_{=0.10} + \underbrace{\text{Cov}_L}_{=0.01} + \underbrace{\text{Cov}_K}_{=0.01}.$$

The first two terms are the sufficient-statistic benchmarks from Proposition 1.1, evaluated using aggregate capital shares and the labor-income-weighted average of the reported sectoral-equivalent elasticities ϕ_f . The remaining terms restore the calibrated system in sequence. Cov_L replaces the scalar labor elasticity with the education-by-occupation supply block and fixed land in a one-capital aggregation. Cov_K then restores the four capital types, the network of investment sectors, their calibrated valuation equations, and the interaction between capital composition and occupation demand. Appendix A.3 derives the reduced long-run system. These are exact sequential corrections—not residuals between dynamic counterfactual models—so the four displayed terms add to the calibrated response of 0.30.

Transition dynamics. What does the model predict in terms of transition dynamics? Figure 6 plots the transition path for prices and quantities. Focusing on the valuation response first, we find that the price response on impact is much larger than in the long run: 1.12 versus 0.30. To tease out which economic force leads to this overshoot, we quantify the benchmark models, and compare their predictions with the calibrated model, as we did for the long run.

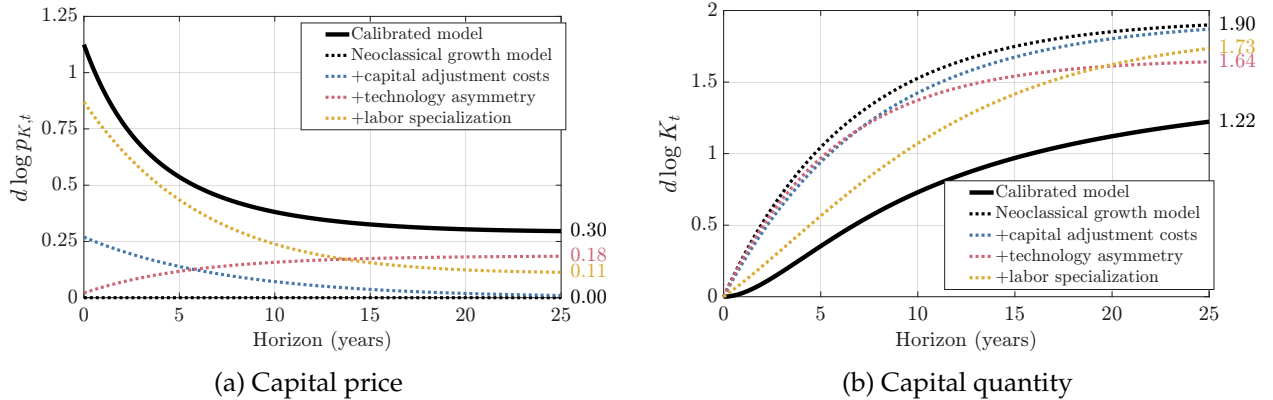


Figure 6: Capital-market transition dynamics

Notes: The calibrated model and special cases are described in Table 4.

We find that both capital adjustment costs and labor specialization matter disproportionately in the short run. Conditional on the steady-state valuation shares, adjustment costs have no independent long-run implication. And for the importance of labor specialization in the short run,

recall that labor supply rises over the horizon, as governed by the parameters $\mu_f, \nu \in (0, 1)$. In contrast, the technology asymmetry force builds slowly over time, alongside capital accumulation. See Table 4 for the short-run responses in these special cases.

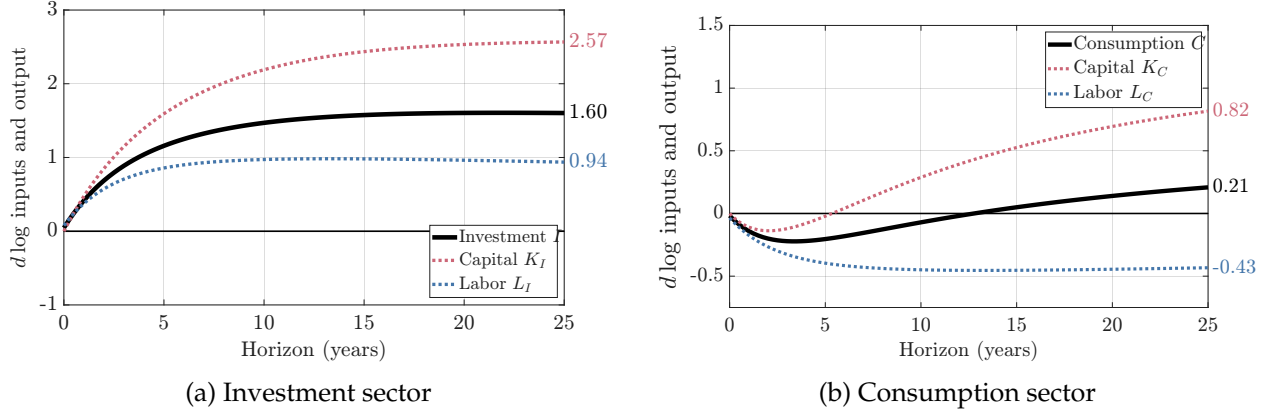


Figure 7: Input reallocation during the transition

Notes: Input types are aggregated using factor shares.

A few remarks on the path of capital accumulation. First, the gap between capital stock in the calibrated model versus in the neoclassical growth model is large throughout the model horizon: it is not just a long-run result. Second, the forces that drive this gap are mostly capital adjustment costs and labor specialization in the short run, and technology asymmetry in the long run. Figure 7 shows the reallocation behind this transition. Capital stocks are predetermined on impact, while labor immediately moves away from consumption and toward investment, raising investment output and producing a temporary decline in consumption. As capital accumulates, capital services in the consumption sector rise; specialized labor nevertheless remains tilted toward investment even in the long run. Consumption crosses its initial level after roughly 13 years, and both sectors ultimately expand. This transition illustrates why a permanently higher desired capital stock need not translate into a large immediate quantity response: the economy must first reallocate scarce inputs, especially specialized labor, toward capital formation.

3.3 Welfare incidence of capital income taxes

We study the welfare incidence of a permanent capital income tax cut evaluated around a distorted steady state with pre-existing capital income tax $\tau \geq 0$. The tax applies to land income as well as to reproducible capital income. The undistorted case $\tau = 0$ is a pure transfer from the government to the household sector; the distorted case $\tau > 0$ additionally generates an efficiency gain. We compute the household welfare effect (i.e., willingness to pay for the tax cut, in units of consumption) using the envelope condition. In the welfare formulas, r denotes the return at the model's quarterly frequency. For a common permanent tax cut $d \log(1 - \tau_{n,t}) = d \log(1 - \tau_{h,t}) = 1$

for every n and all $t \geq 0$, the household-sector welfare gain is

$$\begin{aligned} \mathcal{W} = & \underbrace{(1 - \tau) \left(\sum_n \lambda_{K_n} + \lambda_{L_h} \right) \frac{1+r}{r}}_{\text{direct effect}} + \underbrace{\sum_{t \geq 0} (1+r)^{-t} \sum_n ((1 - \tau) \lambda_{K_n} d \log r_{K_n, t+1} - \lambda_{L_h} d \log p_{L_h, t+1})}_{\text{capital relative-price effects}} \\ & + \underbrace{\sum_{t \geq 0} (1+r)^{-t} \left(\sum_{f \neq h} \lambda_{L_{f_0}} d \log w_{f_0, t+1} + (1 - \tau) \lambda_{L_h} d \log w_{h, t+1} \right)}_{\text{labor and land relative-price effects}}. \end{aligned}$$

The first term is the mechanical tax benefit to reproducible-capital and land owners. The second term captures factor-price effects holding quantities fixed. The factors $1 - \tau$ convert gross factor income into the after-tax income received by owners at the distorted steady state.

The government effect equals the mechanical revenue loss, $-(1 - \tau)(\sum_n \lambda_{K_n} + \lambda_{L_h})(1 + r)/r$, plus τ times the present-value response of the taxed income base. Combining household and government effects gives

$$\mathcal{W} + \mathcal{G} = \tau \lambda_K \sum_{t \geq 0} (1 + r)^{-t} d \log K_{t+1}.$$

Factor-price changes cancel by the linearized unit-cost conditions, and fixed land has no quantity response. Government revenue is valued one-for-one with consumption—equivalently, it is rebated lump sum—so transfers between households and the government cancel in aggregate. At $\tau = 0$, the right-hand side is zero and the tax cut is purely redistributive.¹⁰

The more inelastic aggregate capital is (i.e., the larger the revaluation effect), the more capitalists benefit from the tax cut. Holding discount rates fixed (for instance in a small open economy), capitalists' welfare gain would equal the impact rise in the value of their existing capital stock. When discount rates respond to the tax cut, a wedge opens between the revaluation and the welfare effect (Fagereng et al., 2025), but inelastic capital supply still shifts incidence toward capitalists relative to the neoclassical benchmark. The efficiency gain as the pre-existing tax τ varies is quantified in Figure 8 below.

Results. Table 5 reports welfare incidence at $\tau = 0, 0.20$, and 0.35 , as a share of the net government revenue loss at each tax rate. At $\tau = 0.20$, the calibrated model implies worker and capitalist gains of 0.93 and 0.35 , compared with 1.17 and 0.31 in the neoclassical benchmark. The capitalist-incidence gap is $+4$ pp. Because the tax cut raises efficiency when $\tau > 0$, household gains sum to more than one when divided by the net fiscal cost; the final column nets out the government loss and reports the aggregate Harberger efficiency term. Among capitalists at $\tau = 0.20$, tangible capital owners capture 0.21 , intangible capital owners 0.07 , and landowners 0.07 . Appendix C.2 reports robustness at $\tau = 0.20$.

¹⁰This budget balance also holds with imported intermediates ($\alpha_M > 0$), since import prices are fixed at world levels ($d \log p_M = 0$).

Table 5: Welfare incidence of capital income tax cut

	Workers				Capitalists				Government	Aggregate
	Total	HS	Col.	Post-col.	Total	Tang.	Intan.	Land	Total	Total
<i>Panel A: $\tau = 0$</i>										
Calibrated model	0.65	0.26	0.22	0.17	0.35	0.20	0.09	0.06	-1.00	0.00
Neoclassical growth model	0.75	0.33	0.23	0.19	0.25	0.09	0.07	0.09	-1.00	0.00
+capital adjustment costs	0.73	0.32	0.23	0.19	0.27	0.10	0.08	0.09	-1.00	0.00
+technology asymmetry	0.76	0.33	0.23	0.20	0.24	0.08	0.07	0.09	-1.00	0.00
+labor specialization	0.70	0.30	0.22	0.18	0.30	0.12	0.09	0.09	-1.00	0.00
<i>Panel B: $\tau = 0.20$</i>										
Calibrated model	0.93	0.38	0.31	0.24	0.35	0.21	0.07	0.07	-1.00	0.28
Neoclassical growth model	1.17	0.51	0.36	0.30	0.31	0.11	0.09	0.11	-1.00	0.48
+capital adjustment costs	1.13	0.49	0.35	0.29	0.33	0.12	0.10	0.11	-1.00	0.46
+technology asymmetry	1.14	0.49	0.35	0.30	0.26	0.09	0.07	0.11	-1.00	0.40
+labor specialization	1.07	0.46	0.33	0.28	0.33	0.12	0.10	0.11	-1.00	0.40
<i>Panel C: $\tau = 0.35$</i>										
Calibrated model	1.37	0.56	0.46	0.35	0.34	0.23	0.04	0.08	-1.00	0.71
Neoclassical growth model	2.00	0.87	0.62	0.52	0.43	0.15	0.12	0.16	-1.00	1.43
+capital adjustment costs	1.91	0.83	0.59	0.50	0.45	0.17	0.13	0.15	-1.00	1.37
+technology asymmetry	1.83	0.79	0.56	0.47	0.30	0.09	0.07	0.14	-1.00	1.13
+labor specialization	1.78	0.77	0.55	0.46	0.39	0.14	0.11	0.14	-1.00	1.17

Notes: Welfare changes are divided by the net government revenue loss at the stated tax rate, so the government entry is -1 ; Aggregate sums workers, capitalists, and government. For single-factor benchmarks, within-group incidence is allocated using baseline income shares.

Tables 6 and 7 report annuity-value incidence within workers and capitalists, respectively: wage changes by occupation and post-tax cash-flow changes by capital type.

Table 6: Wage incidence by broad occupation

Occupation	Log wage change (annuity value)	Occupation	Log wage change (annuity value)
Executives and Managers	1.03	Administrative Support	0.61
Engineering and Computing	0.97	Agriculture, Logging, Extraction	0.60
Mechanics and Construction	0.92	Sales, All	0.59
Health and Science Technicians	0.79	Fire, Police, and Guards	0.56
Management Related	0.77	Teachers, Postsecondary	0.51
Scientists	0.73	Food, Cleaning, Personal Services	0.49
Fabricators and Material Handlers	0.70	Doctors and Lawyers	0.48
Precision Manufacturing	0.68	Teachers and Librarians	0.43
Vehicle Operators	0.67	Nurses and Health Services	0.29
Manufacturing Operators	0.66		
Aggregate			0.67

Notes: Annuity values are discounted sums of $d \log w_{0,t}$ multiplied by $r/(1+r)$. Education groups are collapsed using aggregate education wage-bill weights; the 66 market occupations are aggregated to HHJK broad groups, and the aggregate row weights broad occupations, using baseline worker wage-bill shares.

Table 7: Capitalist incidence by capital type

Capital type	Post-tax rental rate (annuity value)	Marginal cost (annuity value)	Cash flow (annuity value)
Land	1.17	–	1.17
Structures	0.61	0.29	0.46
Equipment	0.37	0.18	0.27
IPP	0.47	0.38	0.23
Brand/Org	0.53	0.52	0.21
Aggregate	0.54	0.32	0.35

Notes: Evaluated at $\tau = 0$. Annuity values are discounted sums multiplied by $r/(1+r)$, of $d \log((1 - \tau_n)r_{K_n})$, of $d \log p_{I_n}$, and, for cash flow, of $d \log((1 - \tau_n)r_{K_n}) - \frac{\lambda_{I_n}}{\lambda_{K_n}} d \log p_{I_n}$; the rental-rate and cash-flow columns include the mechanical unit term. Types are ranked by the cash-flow response; the aggregate row weights types by baseline income shares. For land, w_h replaces r_{K_n} .

Harberger triangle. Around a distorted steady state, the marginal efficiency gain from a permanent unit-log tax cut is

$$\tau \lambda_K \sum_{t \geq 0} \left(\frac{1}{1+r} \right)^t d \log K_{t+1}.$$

The formula is the dynamic counterpart of the textbook marginal excess burden: the tax wedge times the induced quantity response. It makes the connection to inelastic capital immediate: if the capital stock barely responds to the tax cut, the efficiency gain is small even when the tax rate is high. Fixed land does not enter because its quantity is unchanged. To second order around $\tau = 0$, eliminating a tax at rate τ changes $\log(1 - \tau)$ by approximately τ . Holding the local capital response fixed and integrating the marginal gain therefore gives the usual Harberger triangle,

$$\frac{1}{2} \tau^2 \lambda_K \sum_{t \geq 0} \left(\frac{1}{1+r} \right)^t d \log K_{t+1}.$$

Figure 8 plots the marginal gain; the shaded area approximates the welfare gain from eliminating the tax.

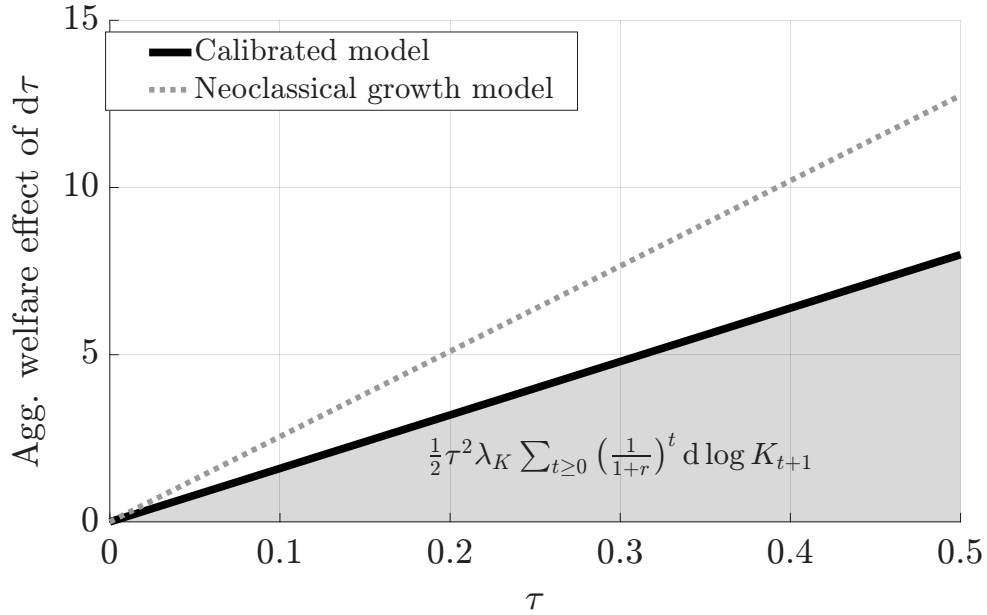


Figure 8: Harberger triangle

Notes: Welfare is expressed in units of annual output; the quarterly discounted sums are divided by four.

3.4 Validation

Investment-demand shocks need not pass entirely into quantities. Table 8 compares the fraction of six nominal responses absorbed by prices or wages with estimates from three empirical studies. Appendix C.3 reports the underlying coefficients and empirical designs.

Table 8: Data and model decompositions of nominal responses

Paper	Capital	Sample	Quantity variable	Price variable	Data	Model
Goolsbee (1998)	Equip.	1962–1988	Equip. output	Output price	0.48	0.80
			Equip. output	Output price	0.31	0.36
House and Shapiro (2008)	Equip. & struct.	2001–2006	Investment purchases	Pre-tax investment price	0.12	0.85
House et al. (2022)	Equip.	1959–2009	Equip. purchases	Equip. purchase price	-0.07	0.66
			Worker hours	Hourly wage	0.40	0.19
	Struct.		Struct. investment	Struct. price	0.38	0.68

For price rows, Data and Model report $d \log p / (d \log Q + d \log p)$, the fraction of the nominal-expenditure response due to prices; the payroll row reports $d \log w / (d \log L + d \log w)$. Values outside $[0, 1]$ arise when the two components move in opposite directions.

Goolsbee (1998) estimate a dynamic IV response across 81 equipment-producing industries; the table reports the current annual coefficient and the cumulative response two years after the price change. House and Shapiro (2008) estimate cross-type WLS regressions for 36 types spanning equipment, quasi-structures, and structures around temporary bonus depreciation; the table maps their structural investment-supply elasticity into the implied total-price share. House et al. (2022) estimate contemporaneous pooled OLS semi-elasticities across equipment or structures types.

4 Conclusion

This paper examines how the labor intensity in the investment sector affects the long-run elasticity of capital supply. In our model, producing investment goods requires specialized labor, making capital supply less responsive to changes in asset prices. Unlike the standard neoclassical model, where capital can be frictionlessly accumulated, our model predicts that the supply of capital becomes inelastic as the share of intangible investment rises.

Using BEA input–output and KLEMS data combined with the Current Population Survey, we document that the investment sector has become more labor-intensive, particularly in terms of college-educated workers. This pattern contrasts with the consumption sector, where the labor share has declined. This rising labor intensity dampens the elasticity of capital supply, making asset prices more sensitive to saving shocks.

We then build and calibrate a realistic multi-sector economy to quantify the supply elasticity of capital. We find that the long-run supply elasticity of capital is finite, around 5, as opposed to the infinite elasticity in the neoclassical model. This lower elasticity shifts tax incidence: capital

owners and investment workers capture a larger share of the benefits of capital tax cuts than in the neoclassical benchmark. As a result, corporate tax cuts disproportionately benefit those involved in capital formation.

References

- Adão, Rodrigo, Martin Beraja, and Nitya Pandalai-Nayar**, “Fast and Slow Technological Transitions,” *Journal of Political Economy Macroeconomics*, 2024, 2 (2), 183–227.
- Atkeson, Andrew and Patrick J Kehoe**, “Modeling and measuring organization capital,” *Journal of Political Economy*, 2005, 113 (5), 1026–1053.
- Auclert, Adrien, Hannes Malmberg, Frédéric Martenet, and Matthew Rognlie**, “Demographics, Wealth, and Global Imbalances in the Twenty-First Century,” Technical Report, National Bureau of Economic Research 2021.
- Baqae, David and Hannes Malmberg**, “Long-Run Comparative Statics,” Technical Report, National Bureau of Economic Research 2025.
- Belo, Frederico, Vito D. Gala, Juliana Salomao, and Maria Ana Vitorino**, “Decomposing Firm Value,” *Journal of Financial Economics*, 2022, 143 (2), 619–639.
- Bernanke, Ben S**, “Global imbalances: recent developments and prospects,” *Bundesbank Lecture*, 2007.
- Böhm, Michael Johannes, Ben Etheridge, and Aitor Irastorza-Fadrique**, “The Impact of Labour Demand Shocks when Occupational Labour Supplies are Heterogeneous,” IZA Discussion Paper 17851, Institute of Labor Economics (IZA) 2025.
- Casal, Lucía and Julieta Caunedo**, “On the Investment Network and Development,” 2024. CEPR Discussion Paper No. 19481.
- Chari, Varadarajan V, Patrick J Kehoe, and Ellen McGrattan**, “The Poverty of Nations: A Quantitative Exploration,” 1996.
- Chodorow-Reich, Gabriel**, “The Neoclassical Theory of Firm Investment and Taxes: A Reassessment,” 2026. NBER Working Paper 33922.
- Corrado, Carol, Charles Hulten, and Daniel Sichel**, “Intangible capital and US economic growth,” *Review of Income and Wealth*, 2009, 55 (3), 661–685.
- , **Jonathan Haskel, Cecilia Jona-Lasinio, and Massimiliano Iommi**, “Intangible capital and modern economies,” *Journal of Economic Perspectives*, 2022, 36 (3), 3–28.
- Crouzet, Nicolas and Janice Eberly**, “Intangibles, markups, and the measurement of productivity growth,” *Journal of Monetary Economics*, 2021, 124, S92–S109.
- **and —**, “Rents and intangible capital: A q+ framework,” *The Journal of Finance*, 2023, 78 (4), 1873–1916.
- , **Janice C Eberly, Andrea L Eisfeldt, and Dimitris Papanikolaou**, “The economics of intangible capital,” *Journal of Economic Perspectives*, 2022, 36 (3), 29–52.
- Cummins, Jason G and Giovanni L Violante**, “Investment-specific technical change in the United States (1947–2000): Measurement and macroeconomic consequences,” *Review of Economic Dynamics*, 2002, 5 (2), 243–284.
- Cummins, Jason G., Kevin A. Hassett, and R. Glenn Hubbard**, “Tax Reforms and Investment: A Cross-Country Comparison,” *Journal of Public Economics*, 1996, 62 (1-2), 237–273.

- Duncan, Otis Dudley and Beverly Duncan**, “A Methodological Analysis of Segregation Indexes,” *American Sociological Review*, 1955, 20 (2), 210–217.
- Eaton, Jonathan and Samuel Kortum**, “Technology, Geography, and Trade,” *Econometrica*, 2002, 70 (5), 1741–1779.
- Eisfeldt, Andrea L and Dimitris Papanikolaou**, “Organization capital and the cross-section of expected returns,” *The Journal of Finance*, 2013, 68 (4), 1365–1406.
- , **Antonio Falato, and Mindy Z Xiaolan**, “Human capitalists,” *NBER Macroeconomics Annual*, 2023, 37 (1), 1–61.
- Fagereng, Andreas, Martin B. Holm, and Gisle J. Natvik**, “MPC Heterogeneity and Household Balance Sheets,” *American Economic Journal: Macroeconomics*, 2021, 13 (4), 1–54.
- , **Matthieu Gomez, Émilien Gouin-Bonenfant, Martin Holm, Benjamin Moll, and Gisle Natvik**, “Asset-Price Redistribution,” *Journal of Political Economy*, 2025, 133 (11), 3494–3549.
- Farhi, Emmanuel and François Gourio**, “Accounting for Macro-Finance Trends: Market Power, Intangibles, and Risk Premia,” *Brookings Papers on Economic Activity*, 2018, pp. 147–250.
- Gomez, Matthieu and Emilien Gouin-Bonenfant**, “How Much Labor Is There in Capital?,” 2025. Working paper.
- Goolsbee, Austan**, “Investment Tax Incentives, Prices, and the Supply of Capital Goods,” *Quarterly Journal of Economics*, 1998, 113 (1), 121–148.
- Greenwood, Jeremy, Zvi Hercowitz, and Per Krusell**, “Long-run Implications of Investment-Specific Technological Change,” *American Economic Review*, 1997, 87 (3), 342–362.
- Gutiérrez, Germán and Thomas Philippon**, “Investmentless Growth: An Empirical Investigation,” *Brookings Papers on Economic Activity*, 2017, pp. 89–169.
- Hayashi, Fumio**, “Tobin’s marginal q and average q : A neoclassical interpretation,” *Econometrica*, 1982, 50 (1), 213–224.
- House, Christopher L, Ana-Maria Mocanu, and Matthew D Shapiro**, “Stimulus Effects of Investment Tax Incentives: Production versus Purchases,” 2022. NBER Working Paper 23391, revised 2022.
- and **Matthew D Shapiro**, “Temporary Investment Tax Incentives: Theory with Evidence from Bonus Depreciation,” *American Economic Review*, 2008, 98 (3), 737–768.
- Hsieh, Chang-Tai, Erik Hurst, Charles I Jones, and Peter J Klenow**, “The Allocation of Talent and US Economic Growth,” *Econometrica*, 2019, 87 (5), 1439–1474.
- Jr, Robert E Lucas**, “Adjustment costs and the theory of supply,” *Journal of Political Economy*, 1967, 75 (4, Part 1), 321–334.
- Justiniano, Alejandro, Giorgio E Primiceri, and Andrea Tambalotti**, “Investment Shocks and the Relative Price of Investment,” *Review of Economic Dynamics*, 2011, 14 (1), 101–121.
- Kaplan, Greg and Giovanni L. Violante**, “A Model of the Consumption Response to Fiscal Stimulus Payments,” *Econometrica*, 2014, 82 (4), 1199–1239.
- Koh, Dongya, Raül Santaaulàlia-Llopis, and Yu Zheng**, “Labor Share Decline and Intellectual Property Products Capital,” *Econometrica*, 2020, 88 (6), 2609–2628.
- Lehn, Christian Vom and Thomas Winberry**, “The Investment Network, Sectoral Comovement, and the Changing US Business Cycle,” *The Quarterly Journal of Economics*, 2022, 137 (1), 387–433.

- McFadden, Daniel**, "The measurement of urban travel demand," *Journal of Public Economics*, 1974, 3 (4), 303–328.
- Mian, Atif R, Ludwig Straub, and Amir Sufi**, "The Saving Glut of the Rich," *Journal of the European Economic Association*, 2024, 22 (3), 1228–1274.
- Mussa, Michael**, "External and internal adjustment costs and the theory of aggregate and firm investment," *Economica*, 1977, 44 (174), 163–178.
- Peters, Ryan H and Lucian A Taylor**, "Intangible capital and the investment-q relation," *Journal of Financial Economics*, 2017, 123 (2), 251–272.
- U.S. Bureau of Economic Analysis**, "Input-Output Accounts Data," 2024. Accessed: 2024-03-20.
- , "KLEMS Industry-Level Productivity Data," 2024. Accessed: 2024-03-20.
- Uzawa, Hirofumi**, "On a two-sector model of economic growth," *The Review of Economic Studies*, 1961, 29 (1), 40–47.
- , "Time preference and the Penrose effect in a two-class model of economic growth," *Journal of Political Economy*, 1969, 77 (4, Part 2), 628–652.
- Valentinyi, Akos and Berthold Herrendorf**, "Measuring factor income shares at the sectoral level," *Review of Economic Dynamics*, 2008, 11 (4), 820–835.

Appendix

A Appendix for Section 1

Table of contents. This appendix collects the theoretical derivations in the following order:

1. **Stylized model:** long-run and transition dynamics.
2. **CES production.**
3. **Multiple capital types.**
4. **Dynamic multi-occupation labor supply.**
5. **Capital adjustment costs and irreversibility.**
6. **Imported inputs.**
7. **Spender-saver discounted Euler.**

A.1 Stylized model: long-run and transition dynamics

Intratemporal block. The stylized model has one reproducible factor K , one labor aggregate L , and two sectors, C and I . We linearize around a steady state with $\tau = 0$. The intratemporal block is

$$\begin{aligned}
 0 &= \alpha_C d \log r_K + (1 - \alpha_C) d \log w_C, \\
 d \log p_K &= \alpha_I d \log r_K + (1 - \alpha_I) d \log w_I, \\
 d \log(r_K K) &= \frac{\alpha_C \lambda_C}{\lambda_K} d \log C + \frac{\alpha_I \lambda_I}{\lambda_K} d \log(p_K I), \\
 d \log C &= \phi d \log \left(\frac{w_C}{w} \right) + d \log w_C, \\
 d \log(p_K I) &= \phi d \log \left(\frac{w_I}{w} \right) + d \log w_I, \\
 d \log w &= \frac{(1 - \alpha_C) \lambda_C}{\lambda_L} d \log w_C + \frac{(1 - \alpha_I) \lambda_I}{\lambda_L} d \log w_I.
 \end{aligned}$$

Given $d \log p_K$ and $d \log K$, this system determines $d \log r_K$, $d \log w_C$, $d \log w_I$, $d \log w$, $d \log C$, and $d \log I$. Solving gives

$$\begin{aligned}
 d \log r_K &= \frac{(1 - \alpha_C) \lambda_I (\alpha_I \lambda_L + \phi (\alpha_I - \alpha_C) \lambda_C)}{\alpha_C (1 - \alpha_I) \lambda_C \lambda_L + \alpha_I (1 - \alpha_C) \lambda_I \lambda_L + \phi \lambda_I \lambda_C (\alpha_I - \alpha_C)^2} d \log p_K \\
 &\quad - \frac{(1 - \alpha_C) (1 - \alpha_I) \lambda_K \lambda_L}{\alpha_C (1 - \alpha_I) \lambda_C \lambda_L + \alpha_I (1 - \alpha_C) \lambda_I \lambda_L + \phi \lambda_I \lambda_C (\alpha_I - \alpha_C)^2} d \log K
 \end{aligned} \tag{A.1}$$

and

$$d \log I = - \frac{\alpha_I}{\alpha_I - \alpha_C} d \log p_K + \frac{(1 - \alpha_C) \lambda_K}{(\alpha_I - \alpha_C) \lambda_I} d \log K + \frac{\lambda_K}{(\alpha_I - \alpha_C) \lambda_I} d \log r_K. \tag{A.2}$$

Substituting (A.1) into (A.2), write

$$d \log r_K = \varepsilon_{r,p} d \log p_K + \varepsilon_{r,K} d \log K, \quad d \log I = \varepsilon_{I,p} d \log p_K + \varepsilon_{I,K} d \log K.$$

Proposition A.1 (Long-run elasticities). *The long-run elasticities of the price and quantity of capital with respect to a capital-demand shock are:*

$$\begin{aligned}\frac{d \log p_K}{d \log(1 - \tau)} &= \underbrace{\frac{\alpha_C - \alpha_I}{1 - \alpha_I}}_{\text{technology asymmetry}} + \underbrace{\frac{1}{1 + \phi} \frac{1 - \alpha_C}{\alpha_C} \frac{\lambda_K}{\lambda_C}}_{\text{labor specialization}}, \\ \frac{d \log K}{d \log(1 - \tau)} &= \frac{1}{\lambda_L} \left(1 - \underbrace{(\alpha_C - \alpha_I) \left(\frac{\lambda_C}{1 - \alpha_I} + \frac{\lambda_I}{\alpha_C} \right)}_{\text{technology asymmetry}} - \underbrace{\frac{1}{1 + \phi} \frac{1 - \alpha_C}{\alpha_C} \lambda_K}_{\text{labor specialization}} \right).\end{aligned}$$

Proof of Proposition 1.1. In steady state, capital accumulation requires $d \log I = d \log K$ and capital valuation requires $d \log p_K = d \log r_K + d \log(1 - \tau)$. Substituting the intratemporal solutions gives the long-run supply and demand curves for capital:

$$\begin{aligned}d \log K &= \frac{\varepsilon_{I,p}}{1 - \varepsilon_{I,K}} d \log p_K && \text{(supply),} \\ d \log p_K &= \frac{\varepsilon_{r,K}}{1 - \varepsilon_{r,p}} d \log K + \frac{1}{1 - \varepsilon_{r,p}} d \log(1 - \tau) && \text{(demand).}\end{aligned}$$

The supply curve collects the (p_K, K) pairs satisfying the intratemporal block together with $I = \delta K$. The demand curve collects the (p_K, K) pairs satisfying the intratemporal block together with $p_K = (1 - \tau)r_K / (r + \delta)$.

Eliminating $d \log K$ and solving:

$$\frac{d \log p_K}{d \log(1 - \tau)} = \frac{1 - \varepsilon_{I,K}}{(1 - \varepsilon_{I,K})(1 - \varepsilon_{r,p}) - \varepsilon_{I,p}\varepsilon_{r,K}}, \quad \frac{d \log K}{d \log(1 - \tau)} = \frac{\varepsilon_{I,p}}{(1 - \varepsilon_{I,K})(1 - \varepsilon_{r,p}) - \varepsilon_{I,p}\varepsilon_{r,K}}.$$

Substituting the explicit formulas for the four intratemporal coefficients from (A.1)–(A.2) yields the closed-form expressions in Proposition 1.1. The key simplification uses

$$\lambda_I(\alpha_C - \alpha_I) + (1 - \alpha_C)\lambda_K = \alpha_C\lambda_L,$$

which follows from $\lambda_K = \alpha_C\lambda_C + \alpha_I\lambda_I$. □

The four intratemporal coefficients can be expressed in terms of the elasticity of investment to capital at fixed prices:

$$\varepsilon_{I,K} = \frac{(1 - \alpha_C)\lambda_K(\alpha_I\lambda_L - (\alpha_C - \alpha_I)\lambda_C\phi)}{\alpha_I(1 - \alpha_C)\lambda_L + (\alpha_C - \alpha_I)\lambda_C\lambda_L + \phi\lambda_I\lambda_C(\alpha_C - \alpha_I)^2}. \quad (\text{A.3})$$

The remaining coefficients are

$$\varepsilon_{I,p} = \frac{\varepsilon_{I,K} - \alpha_I}{\alpha_I - \alpha_C}, \quad \varepsilon_{r,p} = \frac{\lambda_I}{\lambda_K} \varepsilon_{I,K}, \quad \varepsilon_{r,K} = \frac{(\alpha_I - \alpha_C)\lambda_I}{\lambda_K} \varepsilon_{I,K} - (1 - \alpha_C). \quad (\text{A.4})$$

Consumption is pinned down by Hulten's theorem:

$$d \log C = \varepsilon_{C,p} d \log p_K + \varepsilon_{C,K} d \log K, \quad \varepsilon_{C,p} = -\frac{\lambda_I}{\lambda_C} \varepsilon_{I,p}, \quad \varepsilon_{C,K} = \frac{\lambda_K - \lambda_I \varepsilon_{I,K}}{\lambda_C}. \quad (\text{A.5})$$

Proposition 1.1 also gives the exact mapping between price absorption and the dampening of capital

accumulation:

$$\frac{d \log K}{d \log(1 - \tau)} = \frac{1}{\lambda_L} \left(\frac{\lambda_K}{\alpha_C} - \lambda_C \frac{d \log p_K}{d \log(1 - \tau)} \right). \quad (\text{A.6})$$

Factor-price incidence. Unit-cost pricing implies that the weighted change in output prices equals the weighted change in factor prices:

$$\lambda_I d \log p_K = \lambda_K d \log r_K + \lambda_L d \log w, \quad (\text{A.7})$$

where $d \log w$ is the income-share-weighted average of consumption- and investment-sector wage changes. In the long run, capital valuation gives $d \log p_K = d \log r_K + d \log(1 - \tau)$. Substituting into (A.7) yields

$$d \log w = \frac{\lambda_K}{\lambda_L} d \log(1 - \tau) - \frac{\lambda_K - \lambda_I}{\lambda_L} d \log p_K. \quad (\text{A.8})$$

Thus price absorption tilts incidence toward capital holders. The change in the net income earned by capital holders who maintain their stock of capital constant is

$$d \log((1 - \tau)r_K) - \frac{\lambda_I}{\lambda_K} d \log p_K = \frac{\lambda_K - \lambda_I}{\lambda_K} d \log p_K. \quad (\text{A.9})$$

Transition dynamics. Substituting (A.1)–(A.2) into the capital valuation and accumulation equations yields a 2×2 system

$$Ax_{t+1} = Bx_t + Cz_{t+1}, \quad (\text{A.10})$$

where $x_t = (d \log p_{K,t}, d \log K_t)'$ and $z_t = d \log(1 - \tau_t)$. Saddle-path stability requires one eigenvalue of $A^{-1}B$ inside and one outside the unit circle.

Proposition A.2 (Stability condition). *In the non-degenerate two-sector system ($\varepsilon_{I,K} < 1$, which holds whenever $\alpha_C \geq \alpha_I$), the balanced growth path is saddle-path stable if and only if*

$$\frac{\alpha_C - \alpha_I}{1 - \alpha_I} + \frac{1}{1 + \phi} \frac{1 - \alpha_C}{\alpha_C} \frac{\lambda_K}{\lambda_C} > 0. \quad (\text{A.11})$$

Proof. The matrices in (A.10) are

$$A = \begin{pmatrix} \beta(1 - \delta) + \beta(r + \delta)\varepsilon_{r,p} - \frac{\varepsilon_{C,p}}{\psi} & \beta(r + \delta)\varepsilon_{r,K} - \frac{\varepsilon_{C,K}}{\psi} \\ 0 & 1 \end{pmatrix}, \quad (\text{A.12})$$

$$B = \begin{pmatrix} 1 - \frac{\varepsilon_{C,p}}{\psi} & -\frac{\varepsilon_{C,K}}{\psi} \\ \delta\varepsilon_{I,p} & 1 - \delta(1 - \varepsilon_{I,K}) \end{pmatrix}, \quad C = \begin{pmatrix} -\beta(r + \delta) \\ 0 \end{pmatrix}.$$

The first row is the capital valuation equation; the Euler equation contributes the $1/\psi$ terms through (A.5). These terms enter A and B symmetrically, so $A - B$ is independent of ψ .

Using (A.4) and $\lambda_I/\lambda_K = \delta/(r + \delta)$, one can verify that $\det(B) = \det(A)/\beta$. Hence the eigenvalue product is

$$\eta_1 \eta_2 = \frac{\det(B)}{\det(A)} = \frac{1}{\beta} > 1. \quad (\text{A.13})$$

The characteristic polynomial evaluated at $\eta = 1$ equals $\det(A - B)$, which is $-\beta(r + \delta)\delta(1 - \varepsilon_{I,K})$ divided

by the left-hand side of (A.11). Since $\varepsilon_{I,K} < 1$ in the non-degenerate system, the polynomial at $\eta = 1$ is negative exactly when the condition holds, in which case one root lies inside and one outside the unit circle. When the condition fails, both roots are unstable and no bounded equilibrium exists. $\square$

When stability holds, capital is predetermined and prices jump to the saddle path. The impact response is

$$\frac{d \log p_{K,0}}{d \log p_{K,\infty}} = \frac{1 - \eta_s}{\delta(1 - \varepsilon_{I,K})'} \quad (\text{A.14})$$

where $\eta_s \in (0, 1)$ is the stable eigenvalue. To see this, use the accumulation row

$$d \log K_{t+1} = (1 - \delta(1 - \varepsilon_{I,K})) d \log K_t + \delta \varepsilon_{I,p} d \log p_{K,t}. \quad (\text{A.15})$$

Along the saddle path, deviations from the new steady state decay at rate η_s . Matching coefficients on η_s^t and using the long-run supply curve gives (A.14). Asset prices overshoot their long-run level when convergence along the saddle path is faster than capital mean reversion at fixed prices.

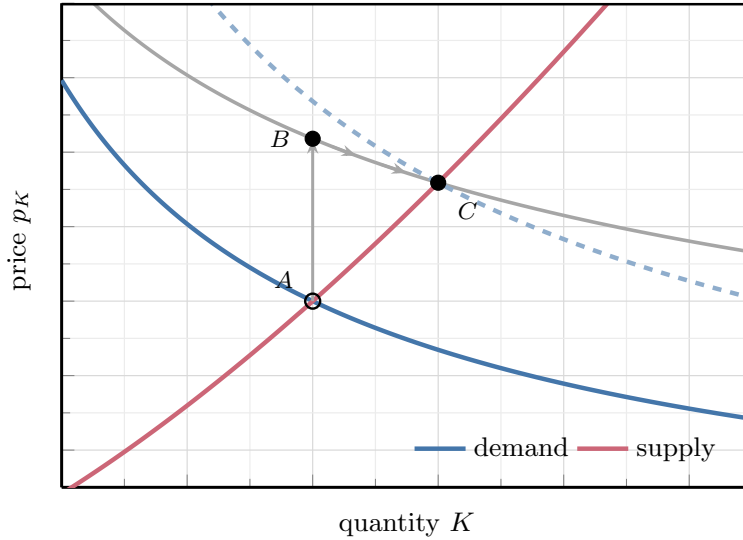


Figure A1: Phase diagram: transition dynamics after a permanent demand shock

A.2 CES production

Start from the stylized model and replace Cobb–Douglas production with CES production in each sector. Let σ_C and σ_I denote the capital–labor elasticities of substitution. The unit-cost equations are unchanged to first order:

$$\begin{aligned} 0 &= \alpha_C d \log r_K + (1 - \alpha_C) d \log w_C, \\ d \log p_K &= \alpha_I d \log r_K + (1 - \alpha_I) d \log w_I. \end{aligned}$$

Only factor demands change:

$$\begin{aligned} d \log K_C &= d \log C + (1 - \alpha_C) \sigma_C d \log \frac{w_C}{r_K}, \\ d \log L_C &= d \log C + \alpha_C \sigma_C d \log \frac{r_K}{w_C}, \\ d \log K_I &= d \log I + (1 - \alpha_I) \sigma_I d \log \frac{w_I}{r_K}, \\ d \log L_I &= d \log I + \alpha_I \sigma_I d \log \frac{r_K}{w_I}. \end{aligned}$$

The labor-supply and intertemporal equations are the same as in the stylized model. Combining CES labor demand with the unchanged Roy labor-supply equations and the unit-cost equations gives the two changed labor-market-clearing equations

$$\begin{aligned} \phi d \log \frac{w_C}{w} + \sigma_C d \log w_C &= d \log C, \\ \phi d \log \frac{w_I}{w} + \sigma_I d \log w_I &= d \log I + \sigma_I d \log p_K. \end{aligned}$$

In the long run, the CES extension has closed-form elasticities:

$$\frac{d \log p_K}{d \log(1 - \tau)} = \frac{\alpha_C - \alpha_I}{1 - \alpha_I} + \frac{1 - \alpha_C}{\alpha_C(\sigma_C + \phi)(1 - \alpha_I)} \left(\sigma_C \alpha_C - \sigma_I \alpha_I \frac{\lambda_K - \lambda_I}{\lambda_C} \right), \quad (\text{A.16})$$

$$\frac{d \log K}{d \log(1 - \tau)} = \sigma_I \frac{\alpha_I}{1 - \alpha_I} + \frac{\phi(1 - \alpha_C) \lambda_C}{\alpha_C(\sigma_C + \phi) \lambda_L(1 - \alpha_I)} \left(\sigma_C \alpha_C - \sigma_I \alpha_I \frac{\lambda_K - \lambda_I}{\lambda_C} \right). \quad (\text{A.17})$$

The Cobb–Douglas case is nested at $\sigma_C = \sigma_I = 1$, in which case (A.16)–(A.17) reduce to Proposition 1.1. Higher substitution in the investment sector raises the quantity response directly. Higher substitution in consumption makes the consumption sector release inputs more elastically, which raises the quantity response; the long-run price response, however, falls with σ_I and rises with σ_C .

A.3 Multiple capital types

Start from the stylized model and allow N capital types and sectors $s \in \{C, I_1, \dots, I_N\}$. The Cobb–Douglas domestic block replaces the two pricing equations with

$$\begin{aligned} 0 &= \sum_{m=1}^N \alpha_{CK_m} d \log r_{K_m} + \sum_f \alpha_{CL_f} d \log w_{Cf}, \\ d \log p_{I_n} &= \sum_{m=1}^N \alpha_{I_n K_m} d \log r_{K_m} + \sum_f \alpha_{I_n L_f} d \log w_{I_n f}. \end{aligned}$$

The corresponding factor-demand equations are the natural multi-factor Cobb–Douglas versions:

$$\begin{aligned} d \log K_{sn} &= d \log Y_s + \sum_{m \neq n} \alpha_{sK_m} d \log \frac{r_{K_m}}{r_{K_n}} + \sum_f \alpha_{sL_f} d \log \frac{w_{sf}}{r_{K_n}}, \\ d \log L_{sf} &= d \log Y_s + \sum_m \alpha_{sK_m} d \log \frac{r_{K_m}}{w_{sf}} + \sum_{g \neq f} \alpha_{sL_g} d \log \frac{w_{sg}}{w_{sf}}. \end{aligned}$$

The CES version uses the same equations after multiplying the capital–labor substitution terms by the relevant σ_s , as in Appendix A.2.

Let

$$\lambda_I = \sum_n \lambda_{I_n}, \quad \alpha_{CK} = \sum_n \alpha_{CK_n}, \quad \alpha_{IK} = \sum_n \frac{\lambda_{I_n}}{\lambda_I} \sum_m \alpha_{I_n K_m},$$

and define aggregate capital-price and capital-quantity responses using capital-income weights,

$$d \log p_K = \sum_n \frac{\lambda_{K_n}}{\lambda_K} d \log p_{K_n}, \quad d \log K = \sum_n \frac{\lambda_{K_n}}{\lambda_K} d \log K_n.$$

For a common long-run shock to all capital types, we decompose the exact calibrated response around the aggregate-share version of Proposition 1.1:

$$\frac{d \log p_K}{d \log(1 - \tau)} = \frac{\alpha_{CK} - \alpha_{IK}}{1 - \alpha_{IK}} + \frac{1}{1 + \sum_f \frac{\lambda_{L_f}}{\lambda_L} \phi_f} \frac{1 - \alpha_{CK}}{\alpha_{CK}} \frac{\lambda_K}{\lambda_C} + \text{Cov}_L + \text{Cov}_K. \quad (\text{A.18})$$

The first two terms are the stylized-model formulas evaluated at the aggregate shares and the labor-income-weighted average of the reported sectoral-equivalent elasticities ϕ_f . The two remaining terms restore the components suppressed by that benchmark in a fixed order, which makes (A.18) an identity rather than an approximation.

First collapse the investment sectors and capital types to one capital type, while preserving the aggregate consumption and investment occupation factor shares. Impose the internally consistent scalar valuation relation from Proposition 1.1, but solve the full education-by-occupation supply block rather than replacing it by the labor-income-weighted average of ϕ_f . The labor correction is the resulting capital-price response less the first two terms:

$$\begin{aligned} \text{Cov}_L \equiv & \left. \frac{d \log p_K}{d \log(1 - \tau)} \right|_{N=1, \text{ occupation block}} - \frac{\alpha_{CK} - \alpha_{IK}}{1 - \alpha_{IK}} \\ & - \frac{1}{1 + \sum_f \frac{\lambda_{L_f}}{\lambda_L} \phi_f} \frac{1 - \alpha_{CK}}{\alpha_{CK}} \frac{\lambda_K}{\lambda_C}. \end{aligned} \quad (\text{A.19})$$

Thus this correction includes the exact occupation matrices, education supply, and fixed land rather than a scalar approximation by education.

Next restore all capital types and investment sectors. Long-run accumulation and investment q imply $d \log K_n = d \log I_n$ and $d \log p_{K_n} = d \log p_{I_n}$. Let $\lambda_{W_n} = \sum_s \lambda_{W_{sn}}$ denote the steady-state wealth share for capital type n . The calibrated valuation equation then gives

$$d \log r_{K_n} = \left(\frac{\lambda_{I_n}}{\lambda_{K_n}} + \frac{r}{1 + r} \frac{\lambda_{W_n}}{\lambda_{K_n}} \right) d \log p_{I_n} - d \log(1 - \tau). \quad (\text{A.20})$$

Conditional on $(\lambda_{K_n}, \lambda_{I_n}, \lambda_{W_n})$, adjustment costs have no independent long-run effect. Using the steady-state identity $\lambda_{W_n} = (1 + r)\lambda_{K_n} / [r + (1 - \theta_n)\delta_n]$ recovers the equivalent coefficient $\lambda_{I_n} / \lambda_{K_n} + r / [r + (1 - \theta_n)\delta_n]$. Thus changing θ_n while recomputing λ_{W_n} changes the steady-state linearization point rather than only the adjustment speed. The coefficient equals one when the scalar model's steady-state user-cost identity holds; the quantitative calibration retains the empirical investment, capital-income, and wealth shares separately.

For Cobb–Douglas production, capital demand and market clearing therefore imply, for every capital type n ,

$$\begin{aligned} d \log I_n &= \frac{\alpha_{CK_n} \lambda_C}{\lambda_{K_n}} (d \log C - d \log r_{K_n}) \\ &+ \sum_m \frac{\alpha_{I_m K_n} \lambda_{I_m}}{\lambda_{K_n}} (d \log I_m + d \log p_{I_m} - d \log r_{K_n}). \end{aligned} \quad (\text{A.21})$$

Likewise, a worker occupation cell (f, o) satisfies

$$\begin{aligned} d \log L_{fo} &= \frac{\alpha_{CL_{fo}} \lambda_C}{\lambda_{L_{fo}}} (d \log C - d \log w_{fo}) \\ &+ \sum_m \frac{\alpha_{I_m L_{fo}} \lambda_{I_m}}{\lambda_{L_{fo}}} (d \log I_m + d \log p_{I_m} - d \log w_{fo}). \end{aligned} \quad (\text{A.22})$$

Equations (A.20)–(A.22), the sector unit-cost equations, the long-run occupation and education-supply equations, and fixed sectoral land quantities form a square linear system. They are the exact long-run reduction of the dynamic system; transition speeds and adjustment speeds drop out.

The capital correction is the difference between this full solution and the one-capital occupation solution used in (A.19):

$$\text{Cov}_K \equiv \left. \frac{d \log p_K}{d \log(1 - \tau)} \right|_{N \text{ capital types, occupation block}} - \left. \frac{d \log p_K}{d \log(1 - \tau)} \right|_{N=1, \text{ occupation block}}. \quad (\text{A.23})$$

This term restores the capital-type composition, the network of investment sectors, the type-specific valuation coefficients in (A.20), and their interaction with occupation demand. The labels Cov_L and Cov_K are retained for continuity, but they are sequential exact corrections rather than standalone covariance formulas. By construction, all four terms in (A.18) add exactly to the calibrated long-run response.

A.4 Dynamic multi-occupation labor supply

This extension starts from the stylized labor block but replaces the scalar sectoral labor-supply curve with an occupation-transition graph. Fix education group f and suppress f on quantities, prices, and occupation bundles. Let occupations be indexed by $o = 1, \dots, O$, let γ_{so} be sector s 's occupation bundle, and let $\phi_{foo'}$ be the long-run elasticity of occupation- o employment with respect to the occupation- o' wage. The elasticities satisfy

$$\sum_{o'=1}^O \phi_{foo'} = 0 \quad \text{for each } o, \quad \sum_{o=1}^O \lambda_{L_o} \phi_{foo'} = 0 \quad \text{for each } o'.$$

Within each sector, occupation demands are Cobb–Douglas:

$$d \log L_{so} = d \log Y_s + \sum_m \alpha_{sK_m} d \log \frac{r_{K_m}}{w_{so}} + \sum_{o' \neq o} \alpha_{sL_{o'}} d \log \frac{w_{so'}}{w_{so}}.$$

In the baseline occupation model, workers are mobile across sectors within an occupation, so $w_{Co} = w_{I_o} \equiv$

w_o and $p_{L_{Co}} = p_{L_{Io}} \equiv p_{L_o}$. The dynamic occupation equation is

$$d \log L_{o,t+1} = (1 - \mu) d \log L_{o,t} + \mu \sum_{o'=1}^O \phi_{foo'} d \log p_{L_{o'},t}, \quad o = 1, \dots, O, \quad (\text{A.24})$$

where variables are measured relative to their occupation-share-weighted aggregate:

$$\sum_{o=1}^O \lambda_{L_o} d \log L_{o,t} = 0.$$

One occupation equation is redundant under this normalization.

The value of choosing occupation o satisfies

$$d \log p_{L_o,t} = \frac{r + \mu}{1 + r} d \log w_{o,t+1} - d \log(1 + r_{t+1}) + \frac{1 - \mu}{1 + r} d \log p_{L_o,t+1}. \quad (\text{A.25})$$

Financial discounting remains governed by r . The coefficient $1 - \mu$ appears because the payoff from the current occupation choice persists only until the worker can reoptimize. In the long run, (A.25) implies $d \log p_{L_o} = d \log w_o$, and (A.24) becomes

$$d \log L_o = \sum_{o'=1}^O \phi_{foo'} d \log w_{o'}, \quad o = 1, \dots, O. \quad (\text{A.26})$$

Identical occupation bundles imply high sectoral mobility because sectoral reallocation does not require changing occupations. Very different occupation bundles imply lower sectoral mobility because expanding one sector requires moving labor across occupations according to the transition graph. The quantitative model solves (A.24)–(A.25) directly.

A.5 Capital adjustment costs and irreversibility

Start from the capital accumulation equation in the stylized model and replace linear installation with the normalized Cobb-Douglas installation technology

$$K_{n,t+1} = (1 - \delta_n) K_{n,t} + \delta_n^{\theta_n} K_{n,t}^{\theta_n} I_{n,t}^{1-\theta_n}.$$

The normalization keeps the steady-state investment-capital ratio equal to δ_n . The linearized accumulation equation is

$$d \log K_{n,t+1} = (1 - (1 - \theta_n)\delta_n) d \log K_{n,t} + (1 - \theta_n)\delta_n d \log I_{n,t}. \quad (\text{A.27})$$

The investment- q equation is

$$d \log I_{n,t} - d \log K_{n,t} = \frac{1}{\theta_n} (d \log p_{K_{n,t}} - d \log p_{I_{n,t}}). \quad (\text{A.28})$$

This is a log-deviation investment- q equation. Let

$$\lambda_{K_n} = \frac{r_{K_n} K_n}{Y}, \quad \lambda_{I_n} = \frac{p_{I_n} I_n}{Y}, \quad \lambda_{W_n} = (1 + r) \frac{p_{K_n} K_n}{Y} = \frac{1 + r}{r + (1 - \theta_n)\delta_n} \lambda_{K_n}.$$

The capital valuation equation is

$$\begin{aligned} d \log p_{K_n,t} = & \frac{\lambda_{K_n}}{\lambda_{W_n}} (d \log(1 - \tau_{n,t+1}) + d \log r_{K_n,t+1}) - \frac{\lambda_{I_n}}{\lambda_{W_n}} d \log p_{I_n,t+1} \\ & - d \log(1 + r_{t+1}) + \frac{1}{1+r} d \log p_{K_n,t+1}. \end{aligned} \quad (\text{A.29})$$

In the long run, (A.28) gives $d \log p_{K_n} = d \log p_{I_n}$, so adjustment costs affect transition dynamics but not the long-run relative price of installed capital.

For the sector-specific irreversibility extension, each sector s owns its own type- n capital stock:

$$d \log K_{sn,t+1} = (1 - (1 - \theta_n)\delta_n) d \log K_{sn,t} + (1 - \theta_n)\delta_n d \log I_{sn,t}, \quad (\text{A.30})$$

$$d \log I_{sn,t} - d \log K_{sn,t} = \frac{1}{\theta_n} (d \log p_{K_{sn,t}} - d \log p_{I_{sn,t}}). \quad (\text{A.31})$$

Define

$$\lambda_{K_{sn}} = \frac{r_{K_{sn}} K_{sn}}{Y}, \quad \lambda_{I_{sn}} = \frac{\lambda_{K_{sn}}}{\lambda_{K_n}} \lambda_{I_n}, \quad \lambda_{W_{sn}} = \frac{1+r}{r + (1 - \theta_n)\delta_n} \lambda_{K_{sn}}.$$

Then (A.29) applies sector by sector using the sector-specific stock, investment, installed-capital price, rental rate, and value shares defined above.

A.6 Imported inputs

Start from the stylized CES model and add imported inputs M_C and M_I with factor shares α_{CM} and α_{IM} . Let

$$\alpha_{CL} = 1 - \alpha_C - \alpha_{CM}, \quad \alpha_{IL} = 1 - \alpha_I - \alpha_{IM},$$

and normalize the import price by consumption, so $d \log p_M = 0$. Import demand is

$$d \log M_C = d \log C, \quad (\text{A.32})$$

$$d \log M_I = d \log I + \sigma_M d \log p_K. \quad (\text{A.33})$$

The unit-cost equations are

$$\begin{aligned} 0 &= \alpha_C d \log r_K + \alpha_{CL} d \log w_C, \\ d \log p_K &= \alpha_I d \log r_K + \alpha_{IL} d \log w_I. \end{aligned}$$

The domestic factor-demand equations change by the own-price substitution toward imports:

$$\begin{aligned} d \log K_C &= d \log C + \sigma_C \alpha_{CL} d \log \frac{w_C}{r_K} - \alpha_{CM} \sigma_M d \log r_K, \\ d \log L_C &= d \log C + \sigma_C \alpha_C d \log \frac{r_K}{w_C} - \alpha_{CM} \sigma_M d \log w_C, \\ d \log K_I &= d \log I + \sigma_I \alpha_{IL} d \log \frac{w_I}{r_K} - \alpha_{IM} \sigma_M d \log r_K, \\ d \log L_I &= d \log I + \sigma_I \alpha_I d \log \frac{r_K}{w_I} - \alpha_{IM} \sigma_M d \log w_I. \end{aligned}$$

Combining these equations with labor supply gives the two changed labor-market-clearing equations:

$$d \log C = \phi d \log \frac{w_C}{w} + (\sigma_C(1 - \alpha_{CM}) + \alpha_{CM}\sigma_M) d \log w_C, \quad (\text{A.34})$$

$$d \log I + \sigma_I d \log p_K = \phi d \log \frac{w_I}{w} + (\sigma_I(1 - \alpha_{IM}) + \alpha_{IM}\sigma_M) d \log w_I. \quad (\text{A.35})$$

For the long-run formulas, write

$$\tilde{\sigma}_C = \sigma_C(1 - \alpha_{CM}) + \alpha_{CM}\sigma_M, \quad \tilde{\sigma}_I = \sigma_I(1 - \alpha_{IM}) + \alpha_{IM}\sigma_M.$$

With $\lambda_K = \alpha_C \lambda_C + \alpha_I \lambda_I$ and $\lambda_L = \alpha_{CL} \lambda_C + \alpha_{IL} \lambda_I$, define the numerator and denominator

$$\begin{aligned} N_M &= \lambda_C \alpha_C [\alpha_{IL}(\alpha_C + \alpha_{CL})\tilde{\sigma}_C + \phi(\alpha_C \alpha_{IL} - \alpha_I \alpha_{CL})] + \alpha_I \alpha_{CL} (\lambda_I \alpha_{IL} - \lambda_C \alpha_C) \tilde{\sigma}_I, \\ D_M &= \lambda_C \alpha_C [\alpha_{IL}(\alpha_C + \alpha_{CL})\tilde{\sigma}_C + \alpha_{CL}(1 - \alpha_I)\tilde{\sigma}_I + \phi(\alpha_C \alpha_{IL} - \alpha_I \alpha_{CL} + \alpha_{CL})] + \alpha_{CL} \alpha_{IL} (\lambda_I \alpha_I \tilde{\sigma}_I - \lambda_K \sigma_I). \end{aligned}$$

The long-run price and quantity responses are

$$\frac{d \log p_K}{d \log(1 - \tau)} = \frac{N_M}{D_M}, \quad (\text{A.36})$$

$$\frac{d \log K}{d \log(1 - \tau)} = \left[\phi \left(\frac{1 - \alpha_I}{\alpha_{IL}} - \frac{\lambda_I - \lambda_K}{\lambda_L} \right) + \tilde{\sigma}_I \frac{1 - \alpha_I}{\alpha_{IL}} - \sigma_I \right] \frac{d \log p_K}{d \log(1 - \tau)} + \phi \left(\frac{\alpha_I}{\alpha_{IL}} - \frac{\lambda_K}{\lambda_L} \right) + \tilde{\sigma}_I \frac{\alpha_I}{\alpha_{IL}}. \quad (\text{A.37})$$

Setting $\alpha_{CM} = \alpha_{IM} = 0$ gives the CES formulas in (A.16)–(A.17). Imports damp domestic factor-price pressure by letting each sector substitute toward a fixed-price input.

A.7 Spender-saver discounted Euler

This extension combines the hand-to-mouth spender block and the discounted Euler equation. Hand-to-mouth agents consume an ω share of fixed-factor income, including labor and land. Savers own reproducible capital and consume the residual. Let C^S denote saver consumption. Aggregate consumption satisfies

$$\lambda_C d \log C_t = \omega \lambda_F d \log w_t^F + (\lambda_C - \omega \lambda_F) d \log C_t^S, \quad (\text{A.38})$$

where $\lambda_F d \log w_t^F$ is the income-share-weighted fixed-factor price change. Saver consumption follows the discounted Euler equation

$$\rho d \log C_{t+1}^S - d \log C_t^S = \psi d \log(1 + r_{t+1}), \quad \rho \in (0, 1]. \quad (\text{A.39})$$

Thus the usual Euler row is written for C^S , while aggregate consumption C is pinned down by (A.38). The representative-agent model is nested at $\omega = 0$ and $\rho = 1$. The pure spender-saver case has $\rho = 1$, so the long-run real rate is still pinned down by the saver Euler equation and the long-run supply-side elasticities are unchanged. When $\rho < 1$, bounded paths imply

$$d \log C_t^S = -\psi \sum_{j=0}^{\infty} \rho^j d \log(1 + r_{t+j+1}),$$

and in a permanent steady state,

$$d \log(1 + r)_\infty = -\frac{1 - \rho}{\psi} d \log C_\infty^S.$$

Thus ω changes how income gains feed into aggregate demand along the transition, while $\rho < 1$ can also change the long-run real-rate response.

B Appendix for Section 2

B.1 Methodology

This appendix describes the two accounting steps used to take the data to the notation of Appendix A. Factor share accounting constructs the sector factor shares α and aggregate income shares λ . Labor reallocation accounting constructs the sector occupation bundles γ_{sfo} and occupation cross-elasticities $\phi_{foo'}$.

B.1.1 Factor share accounting

Let U denote the commodity $\times$ industry use table and M the industry $\times$ commodity make table from the BEA input–output accounts. Let Ω denote the matrix of direct industry factor shares, with one row per industry and one column per factor. The first step converts industry factor shares into direct commodity factor shares. We write

$$\tilde{\Omega} = D' \Omega,$$

where D maps commodity production to producing industries. Under the common industry-technology assumption,

$$D = M \cdot \text{col}(M)^{-1},$$

where $\text{col}(M)$ is the diagonal matrix of column sums. Under the common commodity-technology assumption, our baseline, we instead use

$$D = \left(M' \cdot \text{col}(M')^{-1} \right)^{-1}.$$

The second step converts direct commodity shares into total-requirements shares. Define direct commodity requirements by

$$R = U \cdot \text{col}(M')^{-1} \cdot D,$$

where $\text{col}(M')$ is the diagonal matrix of industry gross output, so column j gives the intermediate commodity inputs used to produce one dollar of commodity j . Stacking commodity factor shares by row, the total-requirements matrix is

$$\Psi = (I - R')^{-1} \tilde{\Omega}.$$

The transpose appears because factor shares are stored as commodity rows: the total factor content of commodity j equals its direct factor content plus the total factor content embodied in every commodity input used to produce j .

The third step aggregates commodity factor shares to the final-use sectors of the model. Let β_s be sector

s 's vector of commodity expenditure shares, where $s \in \{C, I_1, \dots, I_N\}$. Then

$$\alpha'_s = \beta'_s \Psi.$$

The entries of α_s are the factor shares used in Appendix A: α_{sK_n} for reproducible capital type n , α_{sL_f} for education group f , and the land share when land is included. The aggregate income shares are

$$\lambda_s = \frac{p_s Y_s}{Y}, \quad \lambda_{K_n} = \sum_s \lambda_s \alpha_{sK_n}, \quad \lambda_{L_f} = \sum_s \lambda_s \alpha_{sL_f}.$$

For the occupation block, we further split each education share into occupation cells. Let $\alpha_{sL_{fo}}$ be the factor share paid by sector s to education group f and occupation o . The sector occupation bundle is

$$\gamma_{sfo} = \frac{\alpha_{sL_{fo}}}{\alpha_{sL_f}}, \quad \alpha_{sL_f} = \sum_{o=1}^O \alpha_{sL_{fo}}.$$

These γ_{sfo} are the Cobb–Douglas occupation exponents inside the sector-by-education labor aggregate L_{sf} . The corresponding aggregate occupation income shares are

$$\lambda_{L_{fo}} = \sum_s \lambda_s \alpha_{sL_f} \gamma_{sfo}, \quad \lambda_{L_f} = \sum_{o=1}^O \lambda_{L_{fo}}.$$

B.1.2 Labor reallocation accounting

This section constructs the education-specific occupation cross-elasticities $\phi_{f oo'}$ used in the quantitative model. As in Appendix A.4, occupations are indexed by $o = 1, \dots, O$. In the calibration, $O = 66$ and f indexes high school, college, and post-college workers. The elasticities vary by occupation pair within education group, not by sector; sectors enter through the occupation bundles γ_{sfo} .

We use the CPS ASEC because it reports both a worker's current occupation and the worker's occupation in the previous year. We restrict the sample to prime-age workers with valid ASEC weights, nonmissing education, and valid current and previous-year occupations. We map 1990-basis occupations into the occupation groups of Hsieh et al. (2019). We pool all years from 1976–2023 to construct transition probabilities, separately for all workers and for each education group.

For education group f , let $P_{f, oo'}$ denote the pooled probability that a worker whose previous-year occupation is o reports current occupation o' . Let τ_{fo} denote occupation o 's origin share, using CPS person weights. The transition matrix is not itself an elasticity matrix. Following Böhm et al. (2025), we use pairwise occupation flows as sufficient statistics for the relative own- and cross-elasticities of occupational labor supply. First define the current occupation share implied by transitions,

$$s_{fo} = \sum_{o'=1}^O \tau_{fo'} P_{f, o'o}.$$

The same transition matrix pins down the occupation transition speed used in Section 3. Let

$$S_f = \sum_o \tau_{fo} P_{f, oo}$$

denote the origin-share-weighted one-year stay rate. Under the steady-state identity that non-reoptimizers

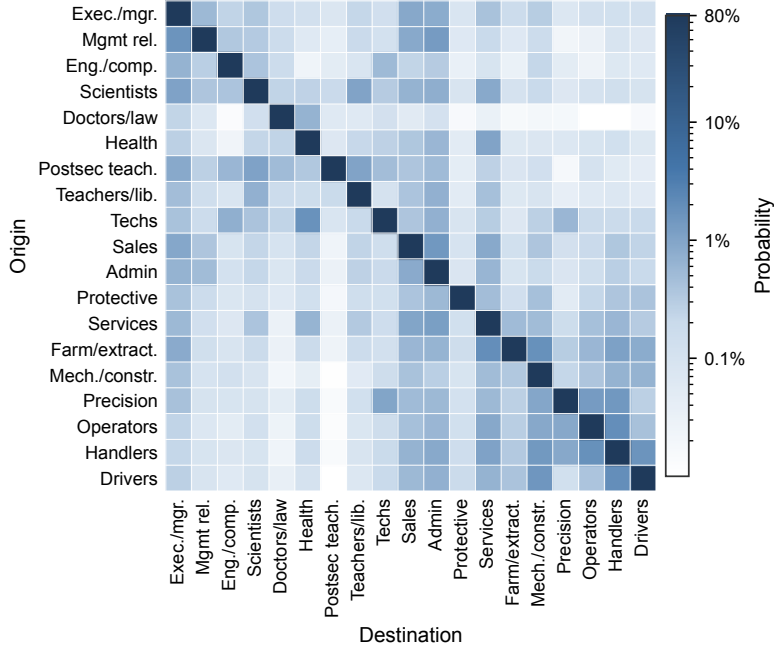


Figure A2: CPS occupation transition matrix

Notes: Rows are previous-year occupations; columns are current occupations. The figure displays HHJK broad groups; the model uses the 66-occupation matrices by education group.

stay in their occupation and reoptimizers redraw from the current occupation distribution s_f , we set

$$\mu_f = \frac{1 - S_f}{1 - \sum_o \tau_{fo} s_{fo}}.$$

Then define the unscaled elasticity element

$$D_{f,oo'} = \frac{1}{\tau_{fo}} \left(\mathbf{1}\{o = o'\} s_{fo} - \sum_{\bar{o}=1}^O \tau_{f\bar{o}} P_{f,\bar{o}o} P_{f,\bar{o}o'} \right).$$

For $o \neq o'$, the second term measures how often occupations o and o' draw from the same origin pool. Hence off-diagonal elements are negative, diagonal elements are positive, and

$$\sum_{o'=1}^O D_{f,oo'} = 0 \quad \text{for each } o, \quad \sum_{o=1}^O \tau_{fo} D_{f,oo'} = 0 \quad \text{for each } o'.$$

A uniform increase in all occupation wages therefore does not reallocate workers across occupations.

The formula pins down the relative pattern of occupational substitutability, but not the overall scale. We choose one common scale factor κ for the occupation classification, using the all-worker matrix so that $\kappa \sum_{o=1}^O \tau_o D_{oo} = 1$. The target is motivated by the Roy–Fréchet occupational sorting benchmark of [Hsieh et al. \(2019, Table II\)](#), whose shape parameter $\bar{\theta} = 2$ corresponds to a unit average own elasticity for the effective-labor object used in our occupation aggregator. We then apply the same κ to each education group. This keeps the occupation-supply scale common across education groups. Differences across education groups come from observed transition patterns, not from choosing a different scale parameter for each

group.

The quantitative model must add up under the model's occupation income shares, not under CPS origin shares. We preserve the transition graph and reweight it to the model shares $\lambda_{L_{f_0}}/\lambda_{L_f}$ by setting

$$\phi_{f_{oo'}} = \kappa \frac{\tau_{f_0}}{\lambda_{L_{f_0}}/\lambda_{L_f}} D_{f,oo'} \quad (o \neq o'), \quad \phi_{f_{00}} = - \sum_{o' \neq 0} \phi_{f_{oo'}}.$$

The resulting model matrix satisfies the same adding-up restrictions as Appendix A.4, now within each education group:

$$\sum_{o'=1}^O \phi_{f_{oo'}} = 0 \quad \text{for each } o, \quad \sum_{o=1}^O \frac{\lambda_{L_{f_0}}}{\lambda_{L_f}} \phi_{f_{oo'}} = 0 \quad \text{for each } o'.$$

The elasticities $\phi_{f_{oo'}}$ and occupation bundles γ_{sf_0} are the labor-side inputs used in the quantitative model. The model does not use a primitive sectoral labor-supply elasticity; it solves directly for occupation-level reallocation within each education group.

For interpretation, we also project the occupation block onto a single consumption-versus-investment margin. Treat aggregate investment as one sector. Its occupation bundle is the wage-bill-weighted average of the four investment-sector bundles:

$$\gamma_{If_0} = \sum_{J \in \mathcal{I}} \frac{\lambda_J \alpha_{JL_f}}{\sum_{J' \in \mathcal{I}} \lambda_{J'} \alpha_{J'L_f}} \gamma_{Jf_0},$$

where $\mathcal{I} = \{I_1, \dots, I_N\}$. Let b_{f_0} be aggregate investment's share of the consumption-plus-investment wage bill in occupation (f, o) :

$$b_{f_0} = \frac{\sum_{J \in \mathcal{I}} \lambda_J \alpha_{JL_f} \gamma_{Jf_0}}{\lambda_C \alpha_{CL_f} \gamma_{Cf_0} + \sum_{J \in \mathcal{I}} \lambda_J \alpha_{JL_f} \gamma_{Jf_0}}.$$

The aggregate investment-versus-consumption sectoral-equivalent elasticity ϕ_f is defined by

$$\frac{1}{1 + \phi_f} = \sum_{o=1}^O \sum_{o'=1}^O (\gamma_{If_0} - \gamma_{Cf_0}) \left[\left(\mathbf{I} + \{\phi_{f_{oo'}}\}_{o,o'=1}^O \right)^{-1} \right]_{oo'} b_{f_{o'}}.$$

Identical occupation bundles imply a small wedge and a high sectoral-equivalent elasticity; very different bundles imply a larger wedge and a lower sectoral-equivalent elasticity. The CPS transition graph refines this comparison by making occupation differences smaller when workers often bridge those occupations in the data.

For robustness, we also construct a transition-weighted analogue of the descriptive dissimilarity measure in the main text. For each education group, define the CPS shared-origin occupation-affinity kernel

$$K_{f,oo'} = \frac{\sum_{\bar{o}=1}^O \tau_{f\bar{o}} P_{f,\bar{o}o} P_{f,\bar{o}o'}}{\left(\sum_{\bar{o}=1}^O \tau_{f\bar{o}} P_{f,\bar{o}o} P_{f,\bar{o}o} \right)^{1/2} \left(\sum_{\bar{o}=1}^O \tau_{f\bar{o}} P_{f,\bar{o}o'} P_{f,\bar{o}o'} \right)^{1/2}}.$$

The transition-weighted occupation distance between consumption and investment type J within education group f is

$$\left[\frac{1}{2} \sum_{o=1}^O \sum_{o'=1}^O (\gamma_{If_0} - \gamma_{Cf_0}) K_{f,oo'} (\gamma_{Jf_{o'}} - \gamma_{Cf_{o'}}) \right]^{1/2}.$$

Table A1: Occupation cross-elasticities by education

Affected occupation o	Wage occupation o'																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
<i>Panel A: HS</i>																			
1. Exec./admin./managerial	93	-7	-2	-3	0	-2	0	-2	-2	-16	-18	-2	-11	-5	-10	-2	-3	-3	-4
2. Management related	-25	126	-4	-3	0	-2	0	-2	-2	-19	-41	-2	-7	-2	-6	-1	-2	-3	-3
3. STEM	-12	-6	87	-5	0	-1	0	-1	-15	-6	-12	-1	-4	-2	-11	-2	-1	-3	-3
4. Science/arts	-15	-5	-5	147	0	-7	0	-13	-8	-17	-21	-2	-27	-3	-9	-2	-4	-5	-4
5. Doctors/lawyers	-75	-18	-12	-24	923	-156	-1	-14	-79	-92	-184	-16	-70	-16	-44	-21	-34	-29	-36
6. Health services	-4	-1	0	-3	-1	96	0	-2	-19	-10	-13	-1	-26	-2	-2	-2	-3	-4	-2
7. Postsec. teachers	-24	-8	-2	-8	-1	-18	215	-12	-13	-14	-26	-5	-32	-5	-14	-4	-9	-12	-9
8. Teachers/librarians	-10	-3	-1	-14	0	-6	-1	122	-3	-14	-26	-2	-22	-2	-5	-2	-4	-4	-3
9. Health/sci. techs.	-6	-2	-7	-4	-1	-26	0	-1	112	-8	-14	-2	-7	-2	-8	-13	-4	-5	-5
10. Sales	-13	-4	-1	-3	0	-4	0	-2	-2	110	-27	-2	-19	-3	-9	-3	-4	-8	-5
11. Admin. support	-9	-6	-1	-2	0	-4	0	-2	-3	-16	76	-1	-13	-2	-4	-2	-3	-6	-4
12. Protective services	-7	-2	-1	-1	0	-3	0	-1	-2	-8	-10	81	-11	-3	-9	-2	-5	-8	-8
13. Food/cleaning/personal	-8	-1	-1	-4	0	-10	0	-3	-2	-17	-18	-2	112	-9	-10	-3	-8	-11	-6
14. Farm/extraction	-11	-1	-1	-2	0	-3	0	-1	-1	-9	-10	-2	-31	144	-28	-5	-10	-16	-14
15. Mechanics/construction	-6	-1	-1	-1	0	-1	0	-1	-2	-7	-5	-2	-9	-7	76	-4	-7	-12	-12
16. Precision mfg.	-5	-1	-1	-1	0	-3	0	-1	-13	-8	-8	-1	-10	-5	-16	126	-23	-26	-5
17. Mfg. operators	-5	-1	0	-1	0	-3	0	-1	-2	-8	-9	-2	-17	-6	-16	-15	128	-34	-8
18. Fabricators/handlers	-4	-1	-1	-1	0	-3	0	-1	-2	-12	-14	-3	-19	-8	-24	-14	-29	165	-28
19. Vehicle operators	-5	-1	-1	-1	0	-1	0	-1	-2	-9	-11	-3	-11	-7	-26	-3	-7	-30	119
<i>Panel B: College</i>																			
1. Exec./admin./managerial	78	-12	-6	-8	-1	-2	-1	-5	-2	-15	-12	-1	-4	-2	-3	-1	-1	-1	-1
2. Management related	-26	89	-6	-7	-1	-1	-1	-4	-2	-15	-18	-1	-3	-1	-2	-1	0	-1	-1
3. STEM	-12	-5	53	-6	-1	0	-1	-2	-9	-4	-5	-1	-2	-1	-3	-1	0	-1	-1
4. Science/arts	-22	-8	-8	114	-1	-4	-1	-18	-6	-11	-13	-1	-12	-1	-2	-1	-2	-1	-1
5. Doctors/lawyers	-138	-96	-64	-70	956	-139	-4	-82	-71	-104	-91	-15	-23	-6	-21	-10	-6	-9	-8
6. Health services	-6	-2	-1	-4	-3	44	-1	-4	-3	-4	-6	0	-7	-1	-1	0	-1	-1	-1
7. Postsec. teachers	-24	-10	-9	-18	-1	-15	194	-34	-22	-15	-19	-1	-11	-3	-6	-2	-3	-1	-1
8. Teachers/librarians	-8	-3	-2	-12	-1	-3	-2	70	-2	-9	-13	-1	-8	-1	-1	-1	-1	-1	-1
9. Health/sci. techs.	-10	-5	-18	-10	-2	-4	-3	-6	101	-7	-12	-1	-5	-1	-3	-8	-2	-2	-3
10. Sales	-23	-10	-3	-6	-1	-2	-1	-7	-2	89	-17	-1	-6	-1	-3	-1	-1	-2	-2
11. Admin. support	-21	-15	-4	-9	-1	-4	-1	-13	-5	-20	113	-2	-8	-1	-2	-1	-1	-3	-2
12. Protective services	-10	-5	-3	-4	-1	-1	0	-6	-3	-8	-10	65	-4	-2	-3	-1	-2	-1	-2
13. Food/cleaning/personal	-22	-7	-4	-25	-1	-14	-2	-22	-6	-21	-25	-2	178	-10	-5	-3	-3	-4	-4
14. Farm/extraction	-33	-7	-5	-10	-1	-4	-1	-14	-5	-14	-14	-3	-32	172	-11	-3	-3	-5	-5
15. Mechanics/construction	-17	-7	-9	-6	-1	-1	-1	-5	-4	-13	-10	-2	-7	-5	107	-4	-3	-6	-7
16. Precision mfg.	-16	-5	-7	-8	-1	-5	-1	-10	-31	-12	-13	-1	-10	-3	-9	175	-17	-23	-2
17. Mfg. operators	-16	-6	-5	-15	-1	-2	-2	-9	-11	-12	-18	-4	-14	-4	-9	-24	204	-42	-8
18. Fabricators/handlers	-14	-7	-7	-9	-1	-3	0	-11	-8	-20	-27	-2	-12	-5	-14	-23	-29	236	-45
19. Vehicle operators	-12	-6	-5	-7	-1	-4	0	-8	-11	-19	-18	-4	-11	-5	-17	-2	-5	-44	179
<i>Panel C: Post-college</i>																			
1. Exec./admin./managerial	72	-12	-6	-11	-6	-2	-4	-9	-2	-9	-6	0	-2	-1	-1	-1	0	0	0
2. Management related	-33	93	-8	-9	-6	-1	-3	-6	-2	-9	-11	-1	-2	-1	-1	0	0	0	0
3. STEM	-13	-6	55	-9	-4	0	-4	-2	-8	-2	-3	0	-1	0	-1	0	0	0	0
4. Science/arts	-19	-6	-8	102	-7	-4	-9	-22	-4	-6	-7	-1	-6	-1	-1	0	0	-1	-1
5. Doctors/lawyers	-8	-3	-3	-5	46	-8	-3	-4	-3	-3	-3	0	0	0	0	0	0	0	0
6. Health services	-5	-1	-1	-7	-17	57	-4	-8	-2	-3	-3	0	-3	0	0	0	0	-1	0
7. Postsec. teachers	-15	-3	-7	-17	-7	-5	92	-18	-4	-4	-6	0	-2	-1	-1	0	-1	-1	0
8. Teachers/librarians	-8	-2	-1	-11	-3	-3	-5	48	-1	-3	-5	0	-3	-1	-1	0	0	0	0
9. Health/sci. techs.	-14	-5	-29	-15	-18	-6	-8	-9	135	-5	-8	-1	-3	0	-3	-5	-2	-2	-2
10. Sales	-32	-12	-4	-12	-7	-4	-4	-13	-2	114	-11	-1	-4	-1	-2	-1	-1	-1	-1
11. Admin. support	-26	-18	-7	-18	-10	-5	-8	-24	-5	-14	152	-1	-6	-1	-2	-1	-1	-2	-2
12. Protective services	-11	-6	-4	-7	-6	-2	-3	-8	-3	-5	-5	72	-4	-1	-3	0	-1	-1	-2
13. Food/cleaning/personal	-33	-8	-7	-43	-5	-14	-8	-50	-6	-14	-19	-3	232	-8	-3	-3	-2	-4	-3
14. Farm/extraction	-39	-10	-4	-28	-9	-5	-13	-28	-3	-19	-9	-2	-27	222	-9	-3	-1	-8	-6
15. Mechanics/construction	-19	-5	-13	-13	-5	-3	-5	-11	-7	-14	-9	-2	-5	-4	137	-2	-2	-9	-8
16. Precision mfg.	-24	-4	-6	-7	-13	-5	-2	-13	-33	-11	-14	-1	-9	-3	-5	184	-17	-17	0
17. Mfg. operators	-29	-11	-9	-16	-3	-6	-13	-18	-16	-12	-13	-2	-7	-2	-6	-27	243	-45	-7
18. Fabricators/handlers	-13	-6	-8	-16	-7	-8	-7	-19	-10	-15	-24	-1	-13	-8	-21	-18	-31	262	-37
19. Vehicle operators	-17	-8	-8	-18	-14	-4	-2	-12	-17	-16	-28	-5	-10	-6	-19	0	-5	-39	229

Notes: Entry (o, o') reports $100 \times \phi_{foo'}$, rounded to the nearest point. Rows are affected occupations; columns are wage occupations, in the same order. The table displays HHJK broad market matrices for readability; the model uses the 66-occupation matrices.

We average these distances across education groups using the relevant consumption-versus-investment wage-bill weights.

B.2 Supplementary material

B.2.1 Robustness tables

Table A2 reports the main robustness checks for end-of-sample average labor shares (2014–2023). Panel A reports total labor shares and Panel B reports college-or-above labor shares. Across specifications, labor shares remain higher in investment than in consumption, though the magnitude of the gap varies.

Table A2: Labor share by sector (2014–2023)

	<i>C</i>	<i>I</i>	<i>I – C</i>
<i>Panel A: Total labor share</i>			
Baseline	0.45	0.59	+0.14
Without capitalizing organizational services	0.51	0.60	+0.09
Capitalizing financial services	0.44	0.56	+0.13
Including government sector	0.49	0.59	+0.10
Cost-shares instead of income-shares	0.62	0.81	+0.19
Using common industry-technology assumption	0.47	0.56	+0.09
<i>Panel B: College (or above) labor share</i>			
Baseline	0.25	0.35	+0.11
Without capitalizing organizational services	0.29	0.33	+0.04
Capitalizing financial services	0.23	0.34	+0.11
Including government sector	0.27	0.35	+0.08
Cost-shares instead of income-shares	0.34	0.48	+0.14
Using common industry-technology assumption	0.26	0.33	+0.07

Without capitalizing organizational services. The first exercise follows the BEA’s current treatment and does not capitalize organizational services. Under this definition, the difference in labor shares across sectors is substantially smaller. This is unsurprising, since the treatment of intangible investment is central to the baseline difference in labor shares between investment and consumption. However, in the calibrated model we allow different types of capital to have different depreciation rates, with intangible capital depreciating faster than structures. From this perspective, organizational services need not be viewed as categorically different from intermediate inputs, but rather as an intermediate form of investment characterized by a higher depreciation rate.

Capitalizing organizational and financial services. Conversely, we consider a broader definition of intangible capital than Corrado et al. (2022), by also capitalizing part of the services in commodity 521CI (“Federal Reserve banks, credit intermediation, and related activities”) and commodity 523 (“Securities and commodity contracts intermediation and brokerage”). The motivation is that these services may reflect costs firms incur—beyond the purchase of physical investment goods—to install productive capital. This alternative classification has little effect on our results, largely because these intermediate expenditures account for less than 3 percent of GDP.

Collapsing private and public sectors. In a third check, we incorporate government activity. As in the private sector, government output can be decomposed into government consumption and government investment. Under this classification, we still find that labor shares are higher in the investment sector, although the gap is smaller because defense spending makes government consumption relatively labor-intensive.

Common industry-technology assumption. To infer commodity-level factor shares from industry-level factor shares, our baseline uses the common commodity-technology assumption, which posits that technology is commodity specific. An alternative is to use the common industry-technology assumption, which posits that technology is industry specific. The factor shares of a commodity can then be constructed as the weighted average of industry-level factor shares, where the weights come from the BEA Make Table. Under this alternative, the labor-share gap between investment and consumption is smaller because commodity-level factor shares smooth out industry-level heterogeneity.

Table A3 compares the summary and detailed BEA classifications. In the main text, we do the analysis at the BEA “summary” level, which includes 71 commodities and is available annually back to 1963. For 2017, the BEA also reports make-use tables using the more detailed 402-commodity classification. The results are very similar, indicating that using the summary classification does not materially affect our findings.

Table A3: Comparing summary and detailed tables

	<i>C</i>	<i>I</i>	Difference
<i>Panel A: Total labor share</i>			
Baseline	0.47	0.59	+0.12
Using detailed input-output tables	0.47	0.58	+0.11
<i>Panel B: College (or above) labor share</i>			
Baseline	0.23	0.33	+0.09
Using detailed input-output tables	0.24	0.32	+0.08

Table A2 reports the robustness checks for end-of-sample average labor shares (2014–2023). Table A4 reports analogous robustness checks for the change in labor shares between the start of the sample (1963–1968) and the end (2014–2023), using the same set of exercises discussed above.

There are two main takeaways. First, essentially all of the increase in the labor share of the investment sector relative to the consumption sector comes from the treatment of organizational services. If organizational services are not capitalized, the investment-consumption labor-share gap is basically stable over time. Second, the other robustness checks do not materially alter the magnitude of the increase in this gap: across specifications, the rise in the investment-consumption labor-share difference ranges from about 5 pp to 10 pp.

B.2.2 Domestic production and imports

The baseline accounting treats imported commodities as if they were produced with the same factor shares as domestically produced commodities. As a robustness exercise, we instead follow the BEA methodology and compute domestic input–output tables, treating imports as a separate input. This separates inputs purchased from abroad from payments to domestic labor and capital.

Table A4: Change in labor income shares between 1963 and 2023 by sector

	ΔC	ΔI	Difference
<i>Panel A: Total labor share</i>			
Baseline	−0.06	+0.00	+0.07
Without capitalizing organization services	−0.03	−0.04	−0.01
Also capitalizing financial services	−0.07	0.00	+0.07
Including government sector	−0.06	−0.01	+0.05
Cost-shares instead of income-shares	−0.06	+0.03	+0.10
Common industry-technology assumption	−0.06	−0.01	+0.06
<i>Panel B: College (or above) labor share</i>			
Baseline	+0.17	+0.27	+0.11
Without capitalizing organization services	+0.20	+0.25	+0.05
Also capitalizing financial services	+0.16	+0.26	+0.11
Including government sector	+0.16	+0.26	+0.10
Cost-shares instead of income-shares	+0.24	+0.37	+0.13
Common industry-technology assumption	+0.17	+0.26	+0.08

We first subtract from the use matrix the import matrix, which reports the use of imports by industries and final uses. We construct this import matrix by assuming that, for each commodity, each use of the imported commodity is proportional to the total use reported in the use table.

Table A5 reports the resulting sector shares, splitting domestic factor shares and imports. Imports account for a large part of equipment investment, but very little of structures, IPP, and organizational investment.

Table A5: Domestic factor shares by sector (2017–2023)

Sector	Factor shares								
	Labor			Land	Capital				Import
	HS	College	Post-college		Equipment	Structures	IPP	Brand/Org	
$C + I$	0.19	0.14	0.12	0.03	0.11	0.14	0.07	0.11	0.10
C	0.19	0.12	0.11	0.04	0.11	0.18	0.06	0.11	0.08
I	0.21	0.18	0.13	0.00	0.10	0.05	0.10	0.10	0.14
$I_{\text{Equipment}}$	0.15	0.11	0.07	0.00	0.09	0.04	0.11	0.09	0.34
$I_{\text{Structures}}$	0.35	0.11	0.06	0.00	0.15	0.08	0.04	0.13	0.08
I_{IPP}	0.14	0.25	0.19	0.00	0.07	0.04	0.16	0.08	0.07
$I_{\text{Brand/Org}}$	0.16	0.27	0.23	0.00	0.07	0.04	0.10	0.08	0.06

B.2.3 Sectoral-equivalent labor-supply elasticities

Table A6 reports the scalar aggregate investment-versus-consumption summaries constructed from the labor reallocation accounting above. These scalars are for interpretation; the quantitative model uses the full set of education-specific cross-elasticities $\phi_{f00'}$ and sector occupation bundles γ_{sfo} .

Table A6: Aggregate investment-versus-consumption sectoral-equivalent elasticities

Education group	$\zeta_f = 1/(1 + \phi_f)$	ϕ_f
HS	0.13	6.47
College	0.14	6.04
Post-college	0.24	3.22
All educations	0.16	5.38

Notes: The all-educations row uses the pooled CPS transition graph and pooled occupation bundles.

C Appendix for Section 3

C.1 Transition paths for factor prices by type

Figure A3 reports transition paths for factor-price responses behind the incidence results in Section 3. The first two panels split asset prices and post-tax rental rates by reproducible capital type and land. The third panel reports wage indexes for the four investment sectors and the consumption sector.

C.2 Robustness of welfare incidence

Table A7 reports how the welfare-incidence shares at $\tau = 0.20$ vary under the alternative modified Euler equations, supply-side parameters, and data constructions used in Table 4. The first row reports the calibrated baseline in levels. The remaining rows report signed deviations from that baseline in percentage points.

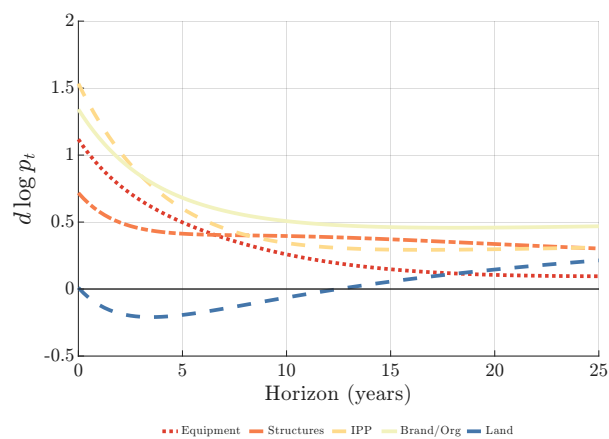
C.3 Validation details

This section reports the published coefficients underlying Table 8 and defines the corresponding empirical and model shares.

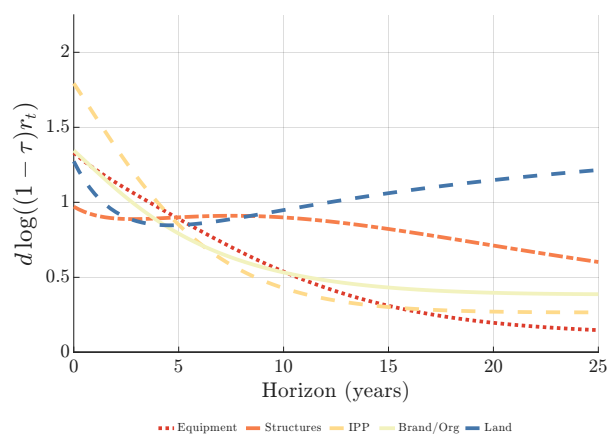
Empirical moments. The “Data” column of Table 8 reports the price share $d \log p / (d \log Q + d \log p)$ for expenditure rows and the wage share $d \log w / (d \log L + d \log w)$ for the payroll row. For Goolsbee (1998), the supply elasticity $\varepsilon_s \equiv d \log Q / d \log p$ implies the price share $1/(1 + \varepsilon_s)$. We use column 6 consistently: the current annual coefficient of 1.09 gives 0.48, while the reported cumulative response two years after the price change, 2.234, gives 0.31.

For House and Shapiro (2008), the structural supply elasticity $\xi = 7.68$ implies that the total pre-tax investment price accounts for $1/(1 + \xi) = 1/(1 + 7.68) = 0.12$ of the nominal-investment response. This price includes external purchase costs and internal adjustment costs. We use the price response implied by the structural quantity equation rather than the separate, noisy observed relative-price coefficient.

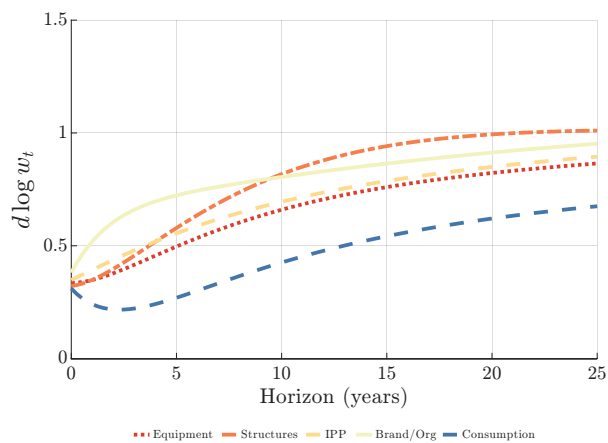
House et al. (2022) estimate common-coefficient pooled OLS semi-elasticities across 28 equipment types and 17 nonresidential structures types, with type-specific linear trends and Driscoll–Kraay standard errors. Equipment purchases and prices are quarterly over 1959:1–2009:4; production inputs are annual over 1959–2009. We use column 1 only, which contains a constant and type-specific linear trends. Its equipment price share is $-0.12/(1.94 - 0.12) = -0.07$. “Production Wages” denotes a production-worker wage bill, so the hourly-wage coefficient is $1.28 - 0.77 = 0.51$ and the wage share of payroll is $(1.28 - 0.77)/1.28 = 0.40$.



(a) Asset prices



(b) Post-tax rental rates



(c) Sector wages

Figure A3: Factor-price transition paths by type

Notes: Land (panels a and b) and the consumption wage (panel c) are shown in blue. Wage indexes are labor-cost-weighted. Post-tax rental rates include the mechanical tax-cut component.

Table A7: Robustness of welfare incidence

	Workers				Capitalists			
	Total	HS	Col.	Post-col.	Total	Tang.	Intan.	Land
Calibrated model	0.93	0.38	0.31	0.24	0.35	0.21	0.07	0.07
Demand-side sensitivities								
intertemporal elasticity $\psi = 0.5$	-7	-3	-2	-2	+5	+4	+1	0
intertemporal elasticity $\psi = 2$	+4	+2	+1	+1	-3	-2	-1	0
spender-saver $\omega = 0.25$	0	0	0	0	0	0	0	0
spender-saver $\omega = 0.50$	-1	0	0	0	0	0	0	0
discount factor $\rho = 0.98$	+5	+2	+1	+1	-3	-2	-1	0
discount factor $\rho = 0.95$	+12	+6	+4	+3	-9	-7	-2	0
Supply-side extensions								
imported inputs $\sigma_M = 3$	+2	0	+1	+1	+4	+3	+1	+1
imported inputs $\sigma_M = 5$	+4	+1	+2	+2	+9	+6	+2	+2
capital-labor elasticity $\sigma_{KL} = 0.5$	-5	-1	-3	-1	-8	-7	-2	0
capital-labor elasticity $\sigma_{KL} = 1.5$	+6	+2	+3	+2	+8	+6	+2	0
occupation transition speeds $2 \times \mu_f$	0	0	0	0	0	0	0	0
occupation transition speeds $10 \times \mu_f$	+1	0	0	0	0	0	0	0
occupation elasticities $2 \times \phi_{f00'}$	0	0	0	0	0	0	0	0
occupation elasticities $0.5 \times \phi_{f00'}$	0	0	0	0	0	0	0	0
Alternative data construction								
no capitalization of org. services	-6	+1	-4	-3	+5	-8	+12	+1
capitalization of financial services	+2	0	+1	+1	0	-13	+12	0
consolidate government sector	+1	0	+1	0	-1	-14	+14	-1
common industry-tech. assumption	-1	0	-1	0	+2	-14	+15	0

Notes: Deviations are in percentage points ($100 \times$ the difference in the share relative to the baseline).

Table A8: Data moments cited in Table 8

Paper	Moment	Source	Value	SE
Goolsbee (1998)	supply elasticity (current annual coefficient)	Table VII, col. 6	1.09	4.15
	supply elasticity (two years after)	Table VII, col. 6, sum of coefficients	2.23	1.32
House and Shapiro (2008)	real-investment supply elasticity ξ	Table 4, Panel A, row 1 (WLS baseline)	7.68	1.85
House et al. (2022)	real equipment purchases	Table 3, col. 1	1.94	0.44
	equipment production price	Table 4, Panel A, col. 1	-0.12	0.10
	production-worker hours	Table 4, Panel A, col. 1	0.77	0.45
	production-worker wage bill	Table 4, Panel A, col. 1	1.28	0.91
	TFP	Table 4, Panel A, col. 1	0.32	0.14
	real structures investment (= production)	Table 4, Panel B, col. 1	0.87	0.32
	structures investment price	Table 4, Panel B, col. 1	0.54	0.12

Structures production equals structures purchases, giving a price share of $0.54 / (0.87 + 0.54) = 0.38$. No structures payroll outcome is available. Cross-equation covariances are not reported, so standard errors for these ratios cannot be recovered from the marginal standard errors.

Empirical designs and model counterparts. The model column holds the 2017–2023 baseline parameterization fixed.

The empirical policies are investment tax incentives. A positive subsidy therefore replaces p_{I_n} by the effective purchase price $p_{I_n} - s_n$ in the investment- q condition and in the investment-price term of the valuation equation. Anticipation and persistence follow the timing of each study.

Goolsbee (1998) use annual data for 81 equipment-producing industries over 1962–1988. The dynamic IV specification instruments the current real output price and two lags with the current tax term and four lags, controlling for industry and year fixed effects, industry trends, and the real wage. The model uses a sustained, unanticipated equipment subsidy. Its current annual counterpart averages quarters 0–3; its two-years-after counterpart averages quarters 8–11.

House and Shapiro (2008) use 36 capital types spanning equipment, quasi-structures, and structures. Type-specific forecasting equations estimated over 1965:1–2000:4 produce real-investment and relative-price forecast errors for 2001:1–2006:4; second-stage WLS identifies the policy effect from cross-type eligibility and depreciation schedules. The initial bonus is anticipated in 2001:IV before its 2002:II effective date; the 50 percent step-up is announced in 2003:II, takes effect in 2003:III, and expires in 2005:I. The model price and quantity paths are projected on this statutory sequence, with the 30 percent phase normalized to one and the 50 percent phase to 5/3; their projected ratio forms the same price-versus-quantity decomposition as $1 / (1 + \xi)$.

The HMS regressions relate current outcomes to the current subsidy. Model moments project the outcomes on an AR(1) subsidy with quarterly persistence 11/12, implying a three-year duration. Wages and hours are annual averages. The payroll counterpart includes all non-land workers rather than only production workers; the structures counterpart applies a direct structures subsidy.

C.4 Adjustment cost, user cost, and capital-type shares

This subsection describes how we calibrate the adjustment cost parameter θ_n , construct the user cost of capital uc_n , and back out time-varying sector-specific capital shares by type. These ingredients feed both the steady-state calibration and the historical decomposition of capital returns.

Adjustment cost. We use BEA Fixed Assets data for depreciation rates by type, averaged over the 2017–2023 period, and set $\delta_{\text{org}} = \delta_{\text{ipp}}$. For type-specific producer prices $p_{I,n}$, we use the producer prices from the same BEA commodity data. Specifically, for each capital type n , we compute the investment-weighted average of commodity-level gross-output price growth factors (propagated through the Leontief inverse), cumulate these into a price level, and normalize to 2017. We construct the consumption price index p_C analogously using private consumption weights. The relative price growth $\pi_{I,n,t}$ is the growth rate of type- n producer price relative to consumption producer price.

We then use a mapping between θ and external evidence on investment-q responses. Let χ denote the quadratic adjustment-cost coefficient, so the slope of the investment rate $p_I I/K$ on Tobin's $q \equiv p_K/p_I$ is $1/\chi$. The quadratic adjustment-cost model predicts the investment semi-elasticity

$$d \left(\frac{p_{I,n,t} I_{n,t}}{K_{n,t}} \right) = \frac{1}{\chi} d \left(\frac{p_{K,n,t}}{p_{I,n,t}} \right)$$

Evaluating in partial equilibrium (i.e., holding p_I constant) and rearranging gives

$$\frac{d \log I_{n,t}}{d \log p_{K,n,t}} = \frac{1}{\chi} \frac{p_{K,n,t} K_{n,t}}{p_{I,n,t} I_{n,t}} \implies \theta = \chi \frac{p_{I,n,t} I_{n,t}}{p_{K,n,t} K_{n,t}}$$

We can further reduce the system using the steady-state q-theory equation $p_I I = (1 - \theta) \delta p_K K$ and we obtain

$$\theta_n = \frac{\delta_n \chi}{1 + \delta_n \chi}.$$

User cost of capital and capital shares. Now, we want to construct an expression for the user cost of capital $uc_n \equiv r + (1 - \theta_n) \delta_n - \pi_{I,n}$. The $(1 - \theta_n)$ factor on depreciation arises because, with the Cobb–Douglas installation function, only a fraction $1 - \theta_n$ of one unit of investment translates into installed capital at the margin; the formula reduces to the textbook $r + \delta - \pi_I$ when $\theta_n = 0$. Identifying sector-specific factor shares by capital type requires assumptions about sector-specific capital values, which cannot be obtained directly from the BEA data; we describe our approach below.

We run the perpetual inventory in quantity units (nominal investment deflated by the type-specific producer price) and accumulate

$$K_{sn,t+1} = (1 - \delta_n) K_{sn,t} + \delta_n^{\theta_n} K_{sn,t}^{\theta_n} I_{sn,t}^{1-\theta_n}.$$

This normalized Cobb–Douglas installation form is the model's internal adjustment-cost technology (Appendix A.5); it reduces to the standard linear PIM, $K_{t+1} = (1 - \delta) K_t + I_t$, when $\theta_n = 0$, and keeps the steady-state investment-capital ratio equal to δ_n . We can then express the capital stock at current prices (relative to consumption) via $K_{sn,t} \cdot p_{I,n,t} / p_{C,t}$. This matches BEA's methodology of building a real quantity index then marking to market. For the initial condition, we use BEA fixed asset stocks (for ORG, which has no BEA counterpart, we initialize with the IPP stock rescaled by the initial ORG-to-IPP investment ratio). As a check on the accumulation procedure, we compare our perpetual-inventory stocks to BEA published fixed-asset stocks at the end of the sample (2022): the PIM-to-BEA ratio is 0.83 for structures, 1.12 for equipment, and 0.96 for IPP. The installation function tracks the BEA stocks to within roughly 20% over the six-decade horizon.

Along a balanced growth path with growth rate g , the aggregate discount rate net of growth is given by

$$r_{t+1} - g = \frac{r_{K,t+1}K_{t+1} - p_{I,t+1}I_{t+1}}{V_t} + \pi_{I,t+1}$$

That is, the aggregate net cash-flow yield on capital (capital income net of investment spending, relative to enterprise value), plus relative investment-price inflation, measures $r - g$. Since the model has no long-run growth, we calibrate the model discount rate to match $r - g$ in the data. For the investment-price inflation term π_I , we use a time-varying weighted average across capital types, weighted by the consumption-equivalent value of each capital type. Both components are smoothed with a five-year backward moving average.

For the capital income yield, we use the nonfinancial corporate (NFC) sector from the IMA, which provides the cleanest measure of capital returns. The enterprise value V_t is NFC equity at market value plus credit market debt and loans. Capital income is NFC gross operating surplus (GOS) net of corporate income taxes. We focus on the NFC sector rather than a consolidated business-sector measure because it avoids imputing mixed income in the noncorporate sector and netting out nonreproducible land from the household sector.

We construct the time-varying sector-specific capital shares by splitting the observed sectoral nonlabor share across capital types in proportion to steady-state capital income implied by observed type-specific investment expenditures:

$$\alpha_{sK_n,t} = \frac{\frac{\bar{r} + \delta_n}{\bar{g} + \delta_n} Ic_{sn,t}}{\sum_m \frac{\bar{r} + \delta_m}{\bar{g} + \delta_m} Ic_{sm,t}} \alpha_{sK,t}, \quad \lambda_{K_n,t} = \sum_s \alpha_{sK_n,t} \lambda_s,$$

where $Ic_{sn,t}$ is sector s 's expenditure on investment of type n , and we set $\bar{r} = 0.05$ and $\bar{g} = 0.025$. In steady state, the capital income of type n equals its investment expenditure scaled by $(\bar{r} + \delta_n)/(\bar{g} + \delta_n)$, so the weights capture the relative capital incomes consistent with observed investment flows. The values $\alpha_{sK,t}$ are observed directly in the BEA (nonlabor share), and the second part of the equation also requires only BEA data (sectoral shares λ_s). Moreover, by construction, we have $\sum_n \lambda_{K_n,t} = \lambda_{K,t}$, where the aggregate capital share is entirely determined by BEA data.